

accumulatedMutations_HSPC

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1 Env

```
rm(list = ls())  
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure5/")
```

2 Libraries

```
library(FSA)
## ## FSA v0.10.0. See citation('FSA') if used in publication.
## ## Run fishR() for related website and fishR('IFAR') for related book.
library(ineq)
library(vegan)
## Loading required package: permute
library(dplyr)
##
## Attaching package: 'dplyr'
## The following objects are masked from 'package:stats':
##
##   filter, lag
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
library(tidyr)
library(ggpubr)
## Loading required package: ggplot2
## Registered S3 methods overwritten by 'car':
##   method      from
##   hist.boot    FSA
##   confint.boot FSA
library(Seurat)
## Loading required package: SeuratObject
## Loading required package: sp
## 'SeuratObject' was built under R 4.5.0 but the current version is
## 4.5.1; it is recommended that you reinstall 'SeuratObject' as the ABI
## for R may have changed
##
## Attaching package: 'SeuratObject'
## The following objects are masked from 'package:base':
##
##   intersect, t
library(igraph)
##
## Attaching package: 'igraph'
## The following object is masked from 'package:Seurat':
##
##   components
## The following object is masked from 'package:tidyr':
##
##   crossing
## The following objects are masked from 'package:dplyr':
##
##   as_data_frame, groups, union
## The following object is masked from 'package:vegan':
##
##   diversity
## The following object is masked from 'package:permute':
##
##   permute
```

```

## The following objects are masked from 'package:stats':
##
##      decompose, spectrum
## The following object is masked from 'package:base':
##
##      union
library(ggraph)
##
## Attaching package: 'ggraph'
## The following object is masked from 'package:sp':
##
##      geometry
library(webshot)
library(ggplot2)
library(ggpmisc)
## Loading required package: ggpp
## Registered S3 methods overwritten by 'ggpp':
##   method                        from
##   heightDetails.titleGrob ggplot2
##   widthDetails.titleGrob  ggplot2
##
## Attaching package: 'ggpp'
## The following objects are masked from 'package:ggpubr':
##
##      as_npc, as_npcx, as_npcy
## The following object is masked from 'package:ggplot2':
##
##      annotate
library(stringr)
library(corrplot)
## corrplot 0.95 loaded
library(pheatmap)
library(reshape2)
##
## Attaching package: 'reshape2'
## The following object is masked from 'package:tidyr':
##
##      smiths
library(tidygraph)
##
## Attaching package: 'tidygraph'
## The following object is masked from 'package:igraph':
##
##      groups
## The following object is masked from 'package:stats':
##
##      filter
library(networkD3)
##
## Attaching package: 'networkD3'
## The following object is masked from 'package:Seurat':
##
##      JS

```

```

## The following object is masked from 'package:SeuratObject':
##
##      JS
library(tidyverse)
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v forcats 1.0.0      v readr  2.1.5
## v lubridate 1.9.4     v tibble  3.3.0
## v purrr     1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x lubridate::%--%()      masks igraph::%--%()
## x ggpp::annotate()      masks ggplot2::annotate()
## x tibble::as_data_frame() masks igraph::as_data_frame(), dplyr::as_data_frame()
## x purrr::compose()      masks igraph::compose()
## x igraph::crossing()     masks tidyr::crossing()
## x tidygraph::filter()   masks dplyr::filter(), stats::filter()
## x dplyr::lag()           masks stats::lag()
## x purrr::simplify()     masks igraph::simplify()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
library(patchwork)

```

3 Functions

```

retriveCellsByLocation <- function(CSM, locList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[, c("CellTag",locList)]
  data_MutPerCell_CSM_tmp <- data.frame(
    CellTag = rownames(CSM_tmp),
    MutNum = rowSums(CSM_tmp == "Mut"),
    WTNNum = rowSums(CSM_tmp == "WT")
  )
  cells_Total <- data_MutPerCell_CSM_tmp[rowSums(data_MutPerCell_CSM_tmp[,c("MutNum", "WTNum")]) > 0, ]
  cells_WT <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "WTNum"] > 0,]$CellTag
  cells_Mut <- data_MutPerCell_CSM_tmp[data_MutPerCell_CSM_tmp[, "MutNum"] > 0,]$CellTag
  cells <- list(Total = cells_Total, Mut = cells_Mut, WT = cells_WT)
  return(cells)
}

retriveLocsByCells <- function(CSM, cellList){
  CSM$CellTag <- rownames(CSM)
  CSM_tmp <- CSM[CSM$CellTag %in% cellList,]
  data_CellPerMut_CSM_tmp <- data.frame(
    Location = colnames(CSM_tmp),
    CellMutNum = colSums(CSM_tmp == "Mut"),
    CellWTNum = colSums(CSM_tmp == "WT")
  )
  pos_Total <- data_CellPerMut_CSM_tmp[rowSums(data_CellPerMut_CSM_tmp[, c("CellMutNum", "CellWTNum")]) > 0, ]
  pos_Mut <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellMutNum"] > 0,]$Location
  pos_WT <- data_CellPerMut_CSM_tmp[data_CellPerMut_CSM_tmp[, "CellWTNum"] > 0,]$Location
  pos <- list(Total = pos_Total, Mut = pos_Mut, WT = pos_WT)
  return(pos)
}

```

```

cal_MultiSpe <- function(matrixData) {

  res <- data.frame(
    Shannon = vegan::diversity(matrixData, index = "shannon"),
    Simpson = vegan::diversity(matrixData, index = "simpson"),
    Gini = apply(matrixData, 1, function(x) ineq(x + 0.00001, type = "Gini")),
    sumMut = rowSums(matrixData),
    Tau = NA
  )

  # Tau = sum(1 - (x / max(x))) / (n - 1)
  res$Tau <- apply(matrixData, 1, function(x) {
    if (all(x == 0)) return(NA)
    x <- x / max(x)
    sum(1 - x) / (length(x) - 1)
  })

  rownames(res) <- rownames(matrixData)
  return(res)
}

scale01 <- function(x) (x - min(x)) / (max(x) - min(x))

cal_scMutScore <- function(data_cell_mut, data_mut_time, data_meta, coefficient_gini = 0.1){
  # data_cell_mut [cell * muts]
  # data_mut_time [muts * tissues/celltype/days]
  # data_meta [Cellbarcode, CellMutTag(tissues/celltype/days)]
  # data_cell_time_matrix
  replacement_map <- c("NoCall" = 0, "WT" = 0, "Mut" = 1)

  data_cell_mut_num <- as.data.frame(lapply(data_cell_mut, function(x) {
    ifelse(x %in% names(replacement_map), replacement_map[x], x)
  })))

  rownames(data_cell_mut_num) <- rownames(data_cell_mut)
  print(dim(data_cell_mut_num))
  print(dim(data_mut_time))

  data_cell_time_matrix <- as.matrix(data_cell_mut_num) %*% as.matrix(data_mut_time)
  rownames(data_cell_time_matrix) <- rownames(data_cell_mut)
  colnames(data_cell_time_matrix) <- colnames(data_mut_time)

  muts_total <- unique(rownames(data_mut_time))

  # get different tissues/time/celltype names
  col_names_data <- colnames(data_cell_time_matrix)

  col_length <- length(col_names_data)

  col_coefficient_fix <- setNames(
    rep(1/col_length, col_length),
    col_names_data
  )
}

```

```

data_cell_time_matrix <- data_cell_time_matrix[,col_names_data]

# Gini
res_index <- cal_MultiSpe(data_cell_time_matrix)

res_diversity <- as.data.frame((data_cell_time_matrix > 1) * rep(col_coefficient_fix[colnames(data_cell_time_matrix)], ncol(data_cell_time_matrix)))
res_diversity$Diversity <- rowSums(res_diversity)

res_diversity$Diversity[is.na(res_diversity$Diversity)] <- 0.00001

res_index$Gini[is.na(res_index$Gini)] <- 0.00001
res_index$Tau[is.na(res_index$Tau)] <- 0.00001

res_index$GiniTau <- res_index$Gini * res_index$Tau

res_index$scMutScore <- 2 / (
  (1 / (coefficient_gini * res_index$GiniTau + 0.00001)) +
  (1 / ((1 - coefficient_gini) * res_diversity$Diversity + 0.00001))
)

res_index$Diversity <- res_diversity$Diversity

res_index$scMutScore <- scales::rescale(res_index$scMutScore, to = c(0, 1))

res <- list(diversity = res_diversity, score = res_index, matrixD = data_cell_time_matrix)
return(res)
}

```

4 LoadData

```

CD34_CSM <- readRDS("../Data/CD34/CD34_CSM.rds")
CD34_SSM <- readRDS("../Data/CD34/CD34_SSM.rds")
CD34_Meta <- readRDS("../Data/CD34/CD34_meta.rds")

```

5 Calculating mutations of each cell and cells of each mutation

```

CD34_MutPerCell <- data.frame(
  CellTag = rownames(CD34_CSM),
  MutNum_somatic = rowSums(CD34_CSM == "Mut"),
  WTNum_somatic = rowSums(CD34_CSM == "WT")
)

CD34_CellPerMut <- data.frame(
  Location = colnames(CD34_CSM),
  CellMutNum = colSums(CD34_CSM == "Mut"),
  CellWTNum = colSums(CD34_CSM == "WT")
)

```

```

cells_candidate <- CD34_MutPerCell[CD34_MutPerCell$MutNum_somatic >= 1,]$CellTag
mut_candidate <- CD34_CellPerMut[CD34_CellPerMut$CellMutNum >= 1,]$Location
CD34_CSM <- CD34_CSM[cells_candidate, mut_candidate]

```

6 calMulti

```

time_types <- c("ery1", "ery2", "ery3", "ery4", "ery5", "ery6", "my1", "my2", "my3", "my4", "prog",

# Cell_Mut
CD34_Meta <- CD34_Meta[CD34_Meta$Cellbarcode %in% cells_candidate,]
CD34_cell_mut_raw <- CD34_CSM

# Mut_Time
CD34_mut_time <- as.data.frame(matrix(nrow = length(mut_candidate), ncol = length(time_types)))
rownames(CD34_mut_time) <- mut_candidate
colnames(CD34_mut_time) <- time_types

for (i in 1:nrow(CD34_mut_time)){
  mut_tmp <- rownames(CD34_mut_time)[i]
  cell_tmp <- retrieveCellsByLocation(CD34_cell_mut_raw, mut_tmp)
  selected_meta_tags <- CD34_Meta[CD34_Meta$Cellbarcode %in% cell_tmp$Mut,]$Group

  CD34_mut_time[i,]$ery1 <- sum(str_count(selected_meta_tags, "ery1"))
  CD34_mut_time[i,]$ery2 <- sum(str_count(selected_meta_tags, "ery2"))
  CD34_mut_time[i,]$ery3 <- sum(str_count(selected_meta_tags, "ery3"))

  CD34_mut_time[i,]$ery4 <- sum(str_count(selected_meta_tags, "ery4"))
  CD34_mut_time[i,]$ery5 <- sum(str_count(selected_meta_tags, "ery5"))
  CD34_mut_time[i,]$ery6 <- sum(str_count(selected_meta_tags, "ery6"))

  CD34_mut_time[i,]$my1 <- sum(str_count(selected_meta_tags, "my1"))
  CD34_mut_time[i,]$my2 <- sum(str_count(selected_meta_tags, "my2"))
  CD34_mut_time[i,]$my3 <- sum(str_count(selected_meta_tags, "my3"))
  CD34_mut_time[i,]$my4 <- sum(str_count(selected_meta_tags, "my4"))

  CD34_mut_time[i,]$prog <- sum(str_count(selected_meta_tags, "prog"))
  CD34_mut_time[i,]$prog_my <- sum(str_count(selected_meta_tags, "prog_my"))
  CD34_mut_time[i,]$prog_ery <- sum(str_count(selected_meta_tags, "prog_ery"))
}

cell_time_num <- sapply(time_types, function(i) sum(str_count(CD34_Meta$Group, i)))
total_cells <- sum(cell_time_num)
cell_time_num_ratio <- cell_time_num/total_cells + 2
CD34_mut_time_nor <- sweep(CD34_mut_time, 2, log2(cell_time_num_ratio), "/")

```

7 computescMutScore

```
res_index_raw_total <- cal_scMutScore(CD34_cell_mut_raw, CD34_mut_time_nor, CD34_Meta, coefficient_gini)
## [1] 1893 478
## [1] 478 13

res_index_raw <- res_index_raw_total$score
res_index_raw$Cellbarcode <- rownames(res_index_raw)

res_matrix <- res_index_raw_total$matrixD

res_index <- merge(res_index_raw, CD34_Meta, by = "Cellbarcode")

selectedGroup <- c("prog", "prog_my", "prog_ery", "my1", "my2", "my3", "my4", "ery1", "ery2", "ery3", "ery4", "ery5")
res_index <- res_index[res_index$Group %in% selectedGroup,]

#res_index$Group[grepl('prog/prog_my/prog_ery/mix', res_index$Group)] <- 'Prog'

res_index$CellMutTag <- factor(res_index$CellMutTag, levels = c("D_8", "D_14", "D_20"))
```

8 checkIndex

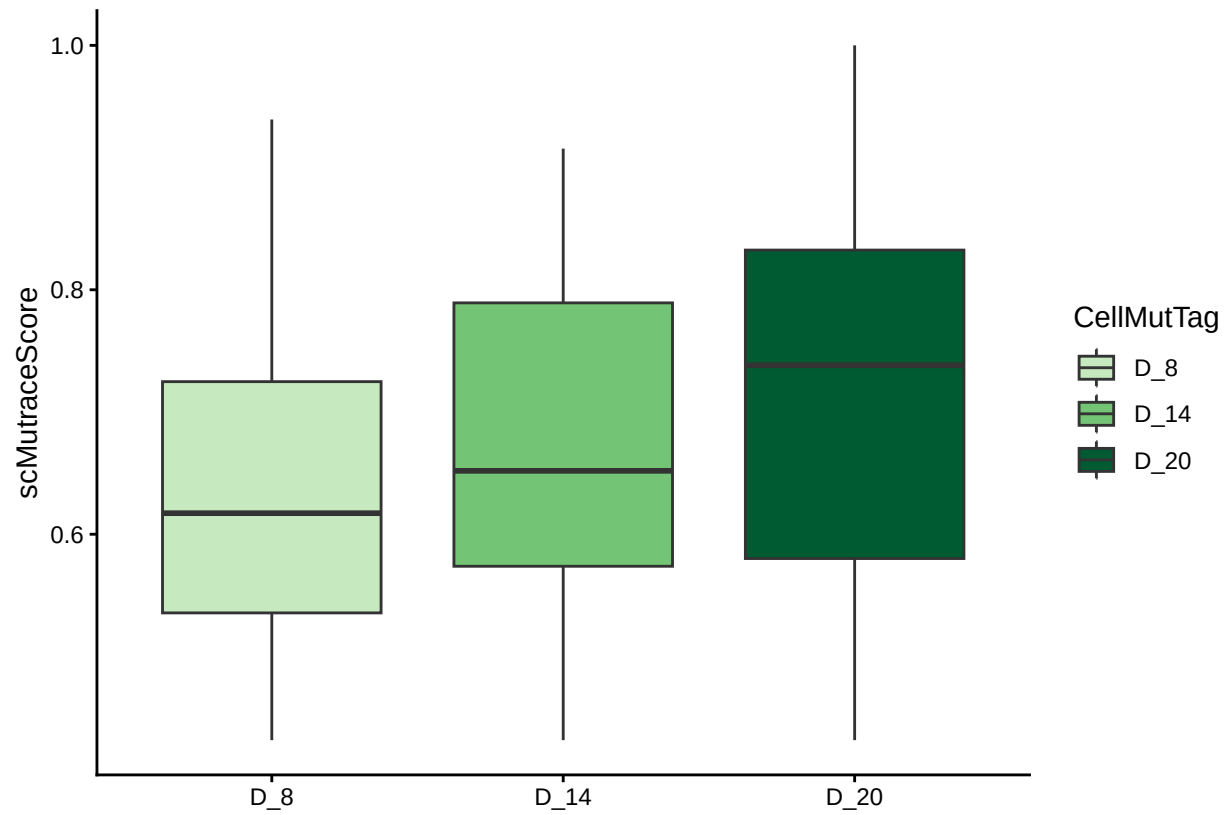
```
cluster_color <- c(my1 = '#6baed6', my2 = '#4292c6', my3 = '#2171b5', my4 = '#084594', prog = '#e0e0e0',
  ery1 = "#bfd3e6", ery2 = "#9ebcda", ery3 = "#8c96c6", ery4 = "#8c6bb1", ery5 = "#88419e",
  prog_ery = "#d8daeb", mix = "#66c2a4", Prog = '#c6dbef', ery1_2 = "#8c96c6", ery3_4 = "#a5d6a7")

day_color <- c(D_8 = "#c7e9c0", D_14 = "#74c476", D_20 = "#005a32")

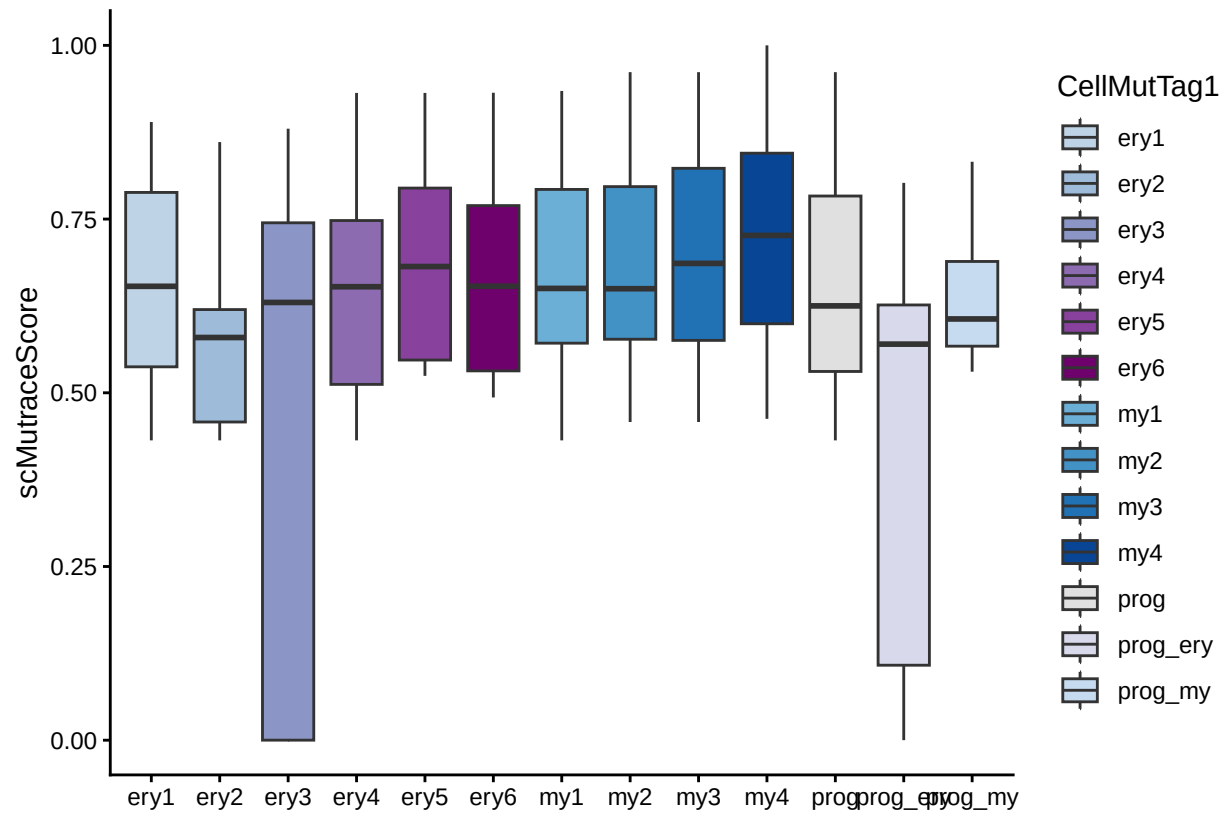
cell_group_color <- c(Prog = "#e5f5f9", Diff = "#66c2a4", Terminal = "#006d2c")

res_index <- res_index[!is.na(res_index$scMutScore),]

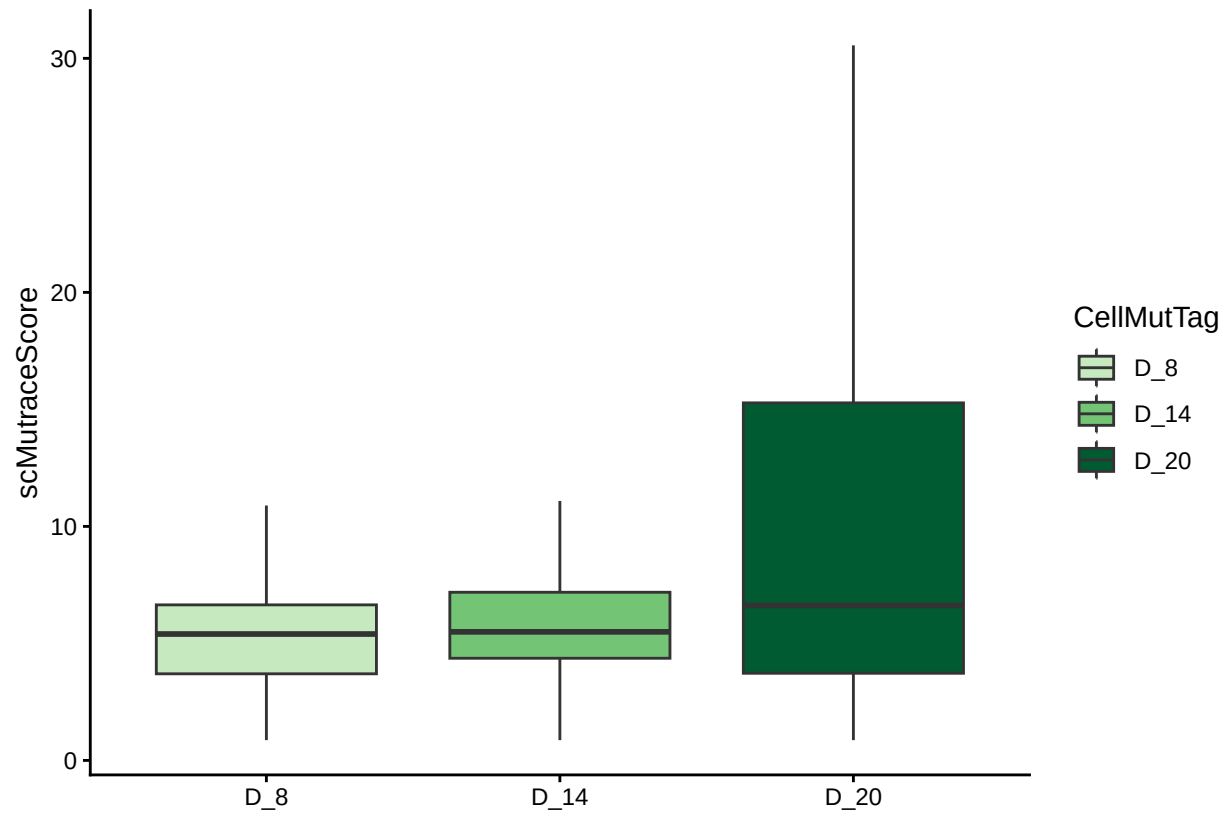
ggplot(data = res_index, mapping = aes(x = CellMutTag, y = scMutScore, fill = CellMutTag)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutScore") +
  scale_fill_manual(values = day_color) +
  theme_classic()
```



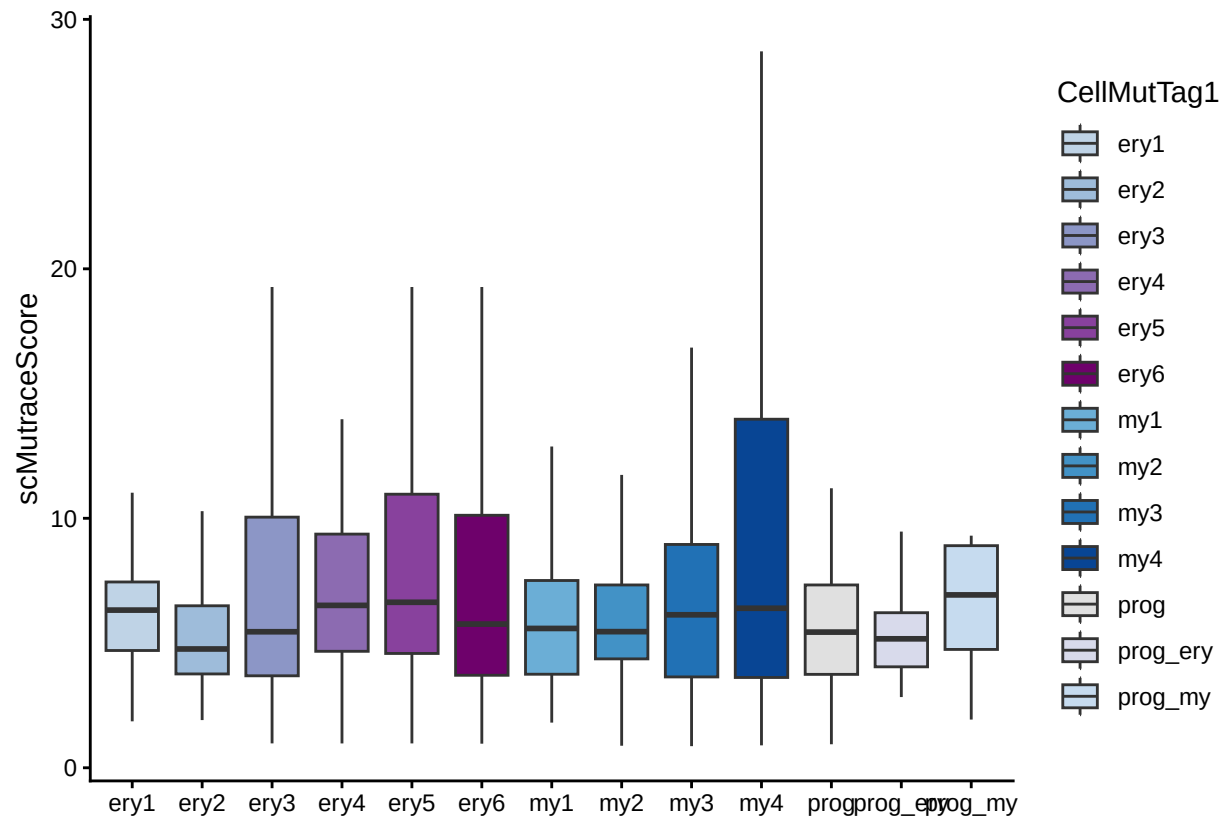
```
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = scMutScore, fill = CellMutTag1)) +  
  geom_boxplot(outliers = FALSE) +  
  labs(x = "", y = "scMutraceScore") +  
  scale_fill_manual(values = cluster_color) +  
  theme_classic()
```



```
ggplot(data = res_index, mapping = aes(x = CellMutTag, y = sumMut, fill = CellMutTag)) +
  geom_boxplot(outliers = FALSE) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = day_color) +
  theme_classic()
```

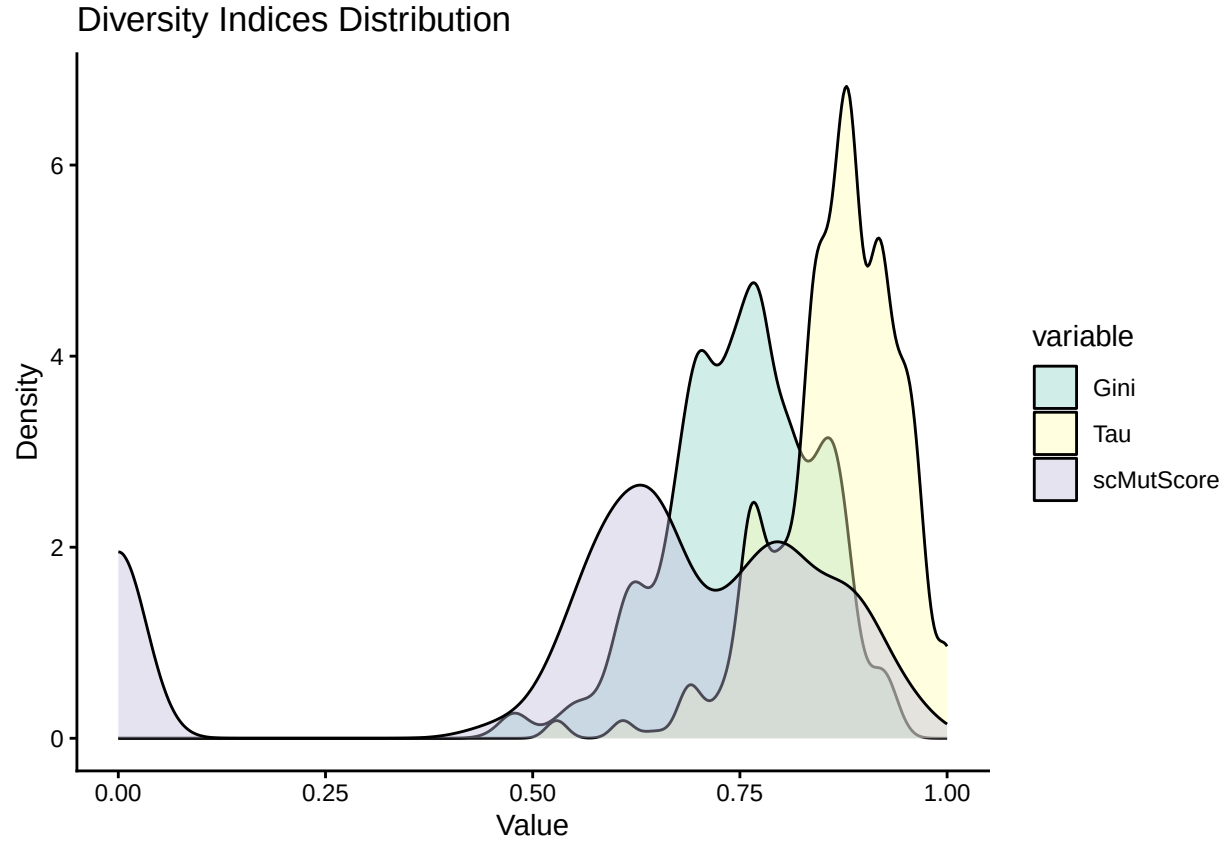


```
ggplot(data = res_index, mapping = aes(x = CellMutTag1, y = sumMut, fill = CellMutTag1)) +  
  geom_boxplot(outliers = FALSE) +  
  labs(x = "", y = "scMutraceScore") +  
  scale_fill_manual(values = cluster_color) +  
  theme_classic()
```



```
res_index_long <- melt(res_index[,c("Gini", "Tau", "scMutScore")])
## No id variables; using all as measure variables

ggplot(res_index_long, aes(x=value, fill=variable)) +
  geom_density(alpha=0.4) +
  theme_classic() +
  labs(x="Value", y="Density", title="Diversity Indices Distribution") +
  scale_fill_brewer(palette="Set3")
```



9 scMutrce validation

```
set.seed(2025)

# 1. init
data_check_CM <- CD34_cell_mut_raw
data_check_Meta <- CD34_Meta
col_check_names <- c("ery1", "ery2", "ery3", "ery4", "ery5", "ery6", "my1", "my2", "my3", "my4", "prog", "mix")
time_types <- c("ery1", "ery2", "ery3", "ery4", "ery5", "ery6", "my1", "my2", "my3", "my4", "prog", "mix")
cluster_color <- c(my1 = '#6baed6', my2 = '#4292c6', my3 = '#2171b5', my4 = '#084594', prog = '#e0e0e0',
  ery1 = "#bfd3e6", ery2 = "#9ebcda", ery3 = "#8c96c6", ery4 = "#8c6bb1", ery5 = "#88419d", ery6 = "#41b6c4",
  prog_ery = "#d8daeb", mix = "#66c2a4", Prog = '#c6dbef', ery1_2 = "#8c96c6", ery3_4 = "#41b6c4")

day_color <- c(D_8 = "#c7e9c0", D_14 = "#74c476", D_20 = "#005a32")

cell_num <- 1
mut_num <- 1
mut_loss_ratio <- 0.5
cell_loss_ratio <- 0.5
iteration_times <- 100

# list for save data
data_scMutModel_list <- vector("list", iteration_times)
```

```

num_cells <- nrow(data_check_CM)
num_mutations <- ncol(data_check_CM)
cells_all <- rownames(data_check_CM)
mut_all <- colnames(data_check_CM)

# 2. loop
for (iter in seq_len(iteration_times)) {

  # 2.1 randomly loss info
  loss_cell_indices <- sample(seq_len(num_cells), ceiling(cell_loss_ratio * num_cells))
  loss_mut_indices <- sample(seq_len(num_mutations), ceiling(mut_loss_ratio * num_mutations))

  C_partial_missing <- data_check_CM
  # new data
  if(length(loss_cell_indices) > 0 && length(loss_mut_indices) > 0){
    C_partial_missing[loss_cell_indices, loss_mut_indices] <- "NoCall"
  }

  # 2.2 count cell/mut number
  MutPerCell <- rowSums(C_partial_missing == "Mut")
  CellPerMut <- colSums(C_partial_missing == "Mut")

  cells_candidate <- cells_all[MutPerCell >= cell_num & cells_all %in% data_check_Meta$Cellbarcode]
  cells_candidate_group <- data_check_Meta[cells_candidate, "Group"]
  muts_candidate <- muts_all[CellPerMut >= mut_num]

  message(paste0("Iteration: ", iter, " loss_cell_indices: ", length(loss_cell_indices), " loss_mut_ind",
  message(paste0("Iteration: ", iter, " cells_candidate: ", length(cells_candidate), " muts_candidate: ")

  data_cell_mut_raw_check <- C_partial_missing[cells_candidate, muts_candidate, drop = TRUE]

  # 2.3 Mut × Time
  cell_time_num <- sapply(col_check_names, function(i) sum(str_count(cells_candidate_group, i)))
  total_cells <- sum(cell_time_num)
  cell_time_ratio <- cell_time_num / total_cells + 2

  # init Mut × Time
  M_partial_missing <- matrix(0, nrow = length(muts_candidate), ncol = length(time_types),
                             dimnames = list(muts_candidate, time_types))

  # Calculate the number of each mutation at each time
  for (t in seq_along(time_types)) {
    time_label <- time_types[t]
    # Find out in which cells each mutation occurs
    mut_cell_list <- apply(data_cell_mut_raw_check, 2, function(x) which(x == "Mut"))
    # Count the number of times each mutation occurs at that point in time
    for (mut in muts_candidate) {
      cells_with_mut <- rownames(data_cell_mut_raw_check)[data_cell_mut_raw_check[, mut] == "Mut"]
      selected_times <- data_check_Meta$Group[data_check_Meta$Cellbarcode %in% cells_with_mut]
      M_partial_missing[mut, t] <- sum(str_count(selected_times, time_label))
    }
  }
}

```

```

# 2.4 normalized Mut × Time
M_partial_missing_norm <- sweep(M_partial_missing, 2, log2(cell_time_ratio), "/")

# 2.5 scMutScore
res_index_tmp <- cal_scMutScore(data_cell_mut_raw_check, M_partial_missing_norm, data_check_Meta, coe
res_index_tmp$iteration <- paste0("iter", iter)
res_index_tmp <- res_index_tmp[res_index_tmp$Diversity > 0, ]
res_index_tmp$Cellbarcode <- rownames(res_index_tmp)
res_index_tmp$iteration <- paste0("iter", iter)
rownames(res_index_tmp) <- NULL
data_scMutModel_list[[iter]] <- res_index_tmp
}

## Iteration: 1 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 1 cells_candidate: 1458 muts_candidate: 451
## [1] 1458 451
## [1] 451 13
## Iteration: 2 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 2 cells_candidate: 1472 muts_candidate: 447
## [1] 1472 447
## [1] 447 13
## Iteration: 3 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 3 cells_candidate: 1465 muts_candidate: 445
## [1] 1465 445
## [1] 445 13
## Iteration: 4 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 4 cells_candidate: 1468 muts_candidate: 447
## [1] 1468 447
## [1] 447 13
## Iteration: 5 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 5 cells_candidate: 1486 muts_candidate: 449
## [1] 1486 449
## [1] 449 13
## Iteration: 6 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 6 cells_candidate: 1427 muts_candidate: 455
## [1] 1427 455
## [1] 455 13
## Iteration: 7 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 7 cells_candidate: 1452 muts_candidate: 450
## [1] 1452 450
## [1] 450 13
## Iteration: 8 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 8 cells_candidate: 1475 muts_candidate: 449
## [1] 1475 449
## [1] 449 13
## Iteration: 9 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 9 cells_candidate: 1476 muts_candidate: 453
## [1] 1476 453
## [1] 453 13
## Iteration: 10 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 10 cells_candidate: 1448 muts_candidate: 446
## [1] 1448 446
## [1] 446 13
## Iteration: 11 loss_cell_indices: 947 loss_mut_indices: 239

```

```

## Iteration: 11 cells_candidate: 1450 muts_candidate: 445
## [1] 1450 445
## [1] 445 13
## Iteration: 12 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 12 cells_candidate: 1427 muts_candidate: 449
## [1] 1427 449
## [1] 449 13
## Iteration: 13 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 13 cells_candidate: 1438 muts_candidate: 452
## [1] 1438 452
## [1] 452 13
## Iteration: 14 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 14 cells_candidate: 1467 muts_candidate: 453
## [1] 1467 453
## [1] 453 13
## Iteration: 15 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 15 cells_candidate: 1471 muts_candidate: 452
## [1] 1471 452
## [1] 452 13
## Iteration: 16 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 16 cells_candidate: 1445 muts_candidate: 451
## [1] 1445 451
## [1] 451 13
## Iteration: 17 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 17 cells_candidate: 1440 muts_candidate: 446
## [1] 1440 446
## [1] 446 13
## Iteration: 18 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 18 cells_candidate: 1460 muts_candidate: 456
## [1] 1460 456
## [1] 456 13
## Iteration: 19 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 19 cells_candidate: 1468 muts_candidate: 442
## [1] 1468 442
## [1] 442 13
## Iteration: 20 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 20 cells_candidate: 1489 muts_candidate: 448
## [1] 1489 448
## [1] 448 13
## Iteration: 21 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 21 cells_candidate: 1465 muts_candidate: 455
## [1] 1465 455
## [1] 455 13
## Iteration: 22 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 22 cells_candidate: 1499 muts_candidate: 447
## [1] 1499 447
## [1] 447 13
## Iteration: 23 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 23 cells_candidate: 1454 muts_candidate: 449
## [1] 1454 449
## [1] 449 13
## Iteration: 24 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 24 cells_candidate: 1498 muts_candidate: 458

```

```

## [1] 1498 458
## [1] 458 13
## Iteration: 25 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 25 cells_candidate: 1430 muts_candidate: 449
## [1] 1430 449
## [1] 449 13
## Iteration: 26 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 26 cells_candidate: 1448 muts_candidate: 448
## [1] 1448 448
## [1] 448 13
## Iteration: 27 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 27 cells_candidate: 1469 muts_candidate: 451
## [1] 1469 451
## [1] 451 13
## Iteration: 28 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 28 cells_candidate: 1458 muts_candidate: 453
## [1] 1458 453
## [1] 453 13
## Iteration: 29 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 29 cells_candidate: 1444 muts_candidate: 453
## [1] 1444 453
## [1] 453 13
## Iteration: 30 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 30 cells_candidate: 1442 muts_candidate: 450
## [1] 1442 450
## [1] 450 13
## Iteration: 31 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 31 cells_candidate: 1476 muts_candidate: 457
## [1] 1476 457
## [1] 457 13
## Iteration: 32 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 32 cells_candidate: 1469 muts_candidate: 455
## [1] 1469 455
## [1] 455 13
## Iteration: 33 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 33 cells_candidate: 1456 muts_candidate: 450
## [1] 1456 450
## [1] 450 13
## Iteration: 34 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 34 cells_candidate: 1464 muts_candidate: 450
## [1] 1464 450
## [1] 450 13
## Iteration: 35 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 35 cells_candidate: 1485 muts_candidate: 451
## [1] 1485 451
## [1] 451 13
## Iteration: 36 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 36 cells_candidate: 1486 muts_candidate: 452
## [1] 1486 452
## [1] 452 13
## Iteration: 37 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 37 cells_candidate: 1441 muts_candidate: 457
## [1] 1441 457

```

```

## [1] 457 13
## Iteration: 38 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 38 cells_candidate: 1464 muts_candidate: 454
## [1] 1464 454
## [1] 454 13
## Iteration: 39 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 39 cells_candidate: 1455 muts_candidate: 452
## [1] 1455 452
## [1] 452 13
## Iteration: 40 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 40 cells_candidate: 1436 muts_candidate: 454
## [1] 1436 454
## [1] 454 13
## Iteration: 41 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 41 cells_candidate: 1483 muts_candidate: 449
## [1] 1483 449
## [1] 449 13
## Iteration: 42 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 42 cells_candidate: 1445 muts_candidate: 458
## [1] 1445 458
## [1] 458 13
## Iteration: 43 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 43 cells_candidate: 1435 muts_candidate: 452
## [1] 1435 452
## [1] 452 13
## Iteration: 44 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 44 cells_candidate: 1433 muts_candidate: 457
## [1] 1433 457
## [1] 457 13
## Iteration: 45 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 45 cells_candidate: 1455 muts_candidate: 455
## [1] 1455 455
## [1] 455 13
## Iteration: 46 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 46 cells_candidate: 1452 muts_candidate: 456
## [1] 1452 456
## [1] 456 13
## Iteration: 47 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 47 cells_candidate: 1427 muts_candidate: 450
## [1] 1427 450
## [1] 450 13
## Iteration: 48 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 48 cells_candidate: 1491 muts_candidate: 436
## [1] 1491 436
## [1] 436 13
## Iteration: 49 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 49 cells_candidate: 1462 muts_candidate: 446
## [1] 1462 446
## [1] 446 13
## Iteration: 50 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 50 cells_candidate: 1461 muts_candidate: 449
## [1] 1461 449
## [1] 449 13

```

```

## Iteration: 51 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 51 cells_candidate: 1450 muts_candidate: 448
## [1] 1450 448
## [1] 448 13
## Iteration: 52 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 52 cells_candidate: 1460 muts_candidate: 444
## [1] 1460 444
## [1] 444 13
## Iteration: 53 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 53 cells_candidate: 1465 muts_candidate: 448
## [1] 1465 448
## [1] 448 13
## Iteration: 54 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 54 cells_candidate: 1428 muts_candidate: 453
## [1] 1428 453
## [1] 453 13
## Iteration: 55 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 55 cells_candidate: 1431 muts_candidate: 443
## [1] 1431 443
## [1] 443 13
## Iteration: 56 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 56 cells_candidate: 1470 muts_candidate: 458
## [1] 1470 458
## [1] 458 13
## Iteration: 57 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 57 cells_candidate: 1465 muts_candidate: 447
## [1] 1465 447
## [1] 447 13
## Iteration: 58 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 58 cells_candidate: 1488 muts_candidate: 443
## [1] 1488 443
## [1] 443 13
## Iteration: 59 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 59 cells_candidate: 1492 muts_candidate: 446
## [1] 1492 446
## [1] 446 13
## Iteration: 60 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 60 cells_candidate: 1462 muts_candidate: 451
## [1] 1462 451
## [1] 451 13
## Iteration: 61 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 61 cells_candidate: 1461 muts_candidate: 452
## [1] 1461 452
## [1] 452 13
## Iteration: 62 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 62 cells_candidate: 1438 muts_candidate: 457
## [1] 1438 457
## [1] 457 13
## Iteration: 63 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 63 cells_candidate: 1441 muts_candidate: 453
## [1] 1441 453
## [1] 453 13
## Iteration: 64 loss_cell_indices: 947 loss_mut_indices: 239

```

```

## Iteration: 64 cells_candidate: 1465 muts_candidate: 449
## [1] 1465 449
## [1] 449 13
## Iteration: 65 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 65 cells_candidate: 1475 muts_candidate: 449
## [1] 1475 449
## [1] 449 13
## Iteration: 66 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 66 cells_candidate: 1469 muts_candidate: 454
## [1] 1469 454
## [1] 454 13
## Iteration: 67 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 67 cells_candidate: 1480 muts_candidate: 453
## [1] 1480 453
## [1] 453 13
## Iteration: 68 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 68 cells_candidate: 1461 muts_candidate: 441
## [1] 1461 441
## [1] 441 13
## Iteration: 69 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 69 cells_candidate: 1461 muts_candidate: 444
## [1] 1461 444
## [1] 444 13
## Iteration: 70 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 70 cells_candidate: 1478 muts_candidate: 450
## [1] 1478 450
## [1] 450 13
## Iteration: 71 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 71 cells_candidate: 1486 muts_candidate: 454
## [1] 1486 454
## [1] 454 13
## Iteration: 72 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 72 cells_candidate: 1430 muts_candidate: 451
## [1] 1430 451
## [1] 451 13
## Iteration: 73 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 73 cells_candidate: 1492 muts_candidate: 452
## [1] 1492 452
## [1] 452 13
## Iteration: 74 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 74 cells_candidate: 1427 muts_candidate: 453
## [1] 1427 453
## [1] 453 13
## Iteration: 75 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 75 cells_candidate: 1469 muts_candidate: 449
## [1] 1469 449
## [1] 449 13
## Iteration: 76 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 76 cells_candidate: 1447 muts_candidate: 448
## [1] 1447 448
## [1] 448 13
## Iteration: 77 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 77 cells_candidate: 1456 muts_candidate: 449

```

```

## [1] 1456 449
## [1] 449 13
## Iteration: 78 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 78 cells_candidate: 1499 muts_candidate: 452
## [1] 1499 452
## [1] 452 13
## Iteration: 79 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 79 cells_candidate: 1477 muts_candidate: 448
## [1] 1477 448
## [1] 448 13
## Iteration: 80 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 80 cells_candidate: 1450 muts_candidate: 448
## [1] 1450 448
## [1] 448 13
## Iteration: 81 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 81 cells_candidate: 1478 muts_candidate: 450
## [1] 1478 450
## [1] 450 13
## Iteration: 82 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 82 cells_candidate: 1439 muts_candidate: 455
## [1] 1439 455
## [1] 455 13
## Iteration: 83 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 83 cells_candidate: 1464 muts_candidate: 460
## [1] 1464 460
## [1] 460 13
## Iteration: 84 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 84 cells_candidate: 1455 muts_candidate: 447
## [1] 1455 447
## [1] 447 13
## Iteration: 85 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 85 cells_candidate: 1455 muts_candidate: 446
## [1] 1455 446
## [1] 446 13
## Iteration: 86 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 86 cells_candidate: 1451 muts_candidate: 441
## [1] 1451 441
## [1] 441 13
## Iteration: 87 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 87 cells_candidate: 1460 muts_candidate: 453
## [1] 1460 453
## [1] 453 13
## Iteration: 88 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 88 cells_candidate: 1467 muts_candidate: 449
## [1] 1467 449
## [1] 449 13
## Iteration: 89 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 89 cells_candidate: 1485 muts_candidate: 451
## [1] 1485 451
## [1] 451 13
## Iteration: 90 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 90 cells_candidate: 1456 muts_candidate: 454
## [1] 1456 454

```

```

## [1] 454 13
## Iteration: 91 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 91 cells_candidate: 1463 muts_candidate: 453
## [1] 1463 453
## [1] 453 13
## Iteration: 92 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 92 cells_candidate: 1471 muts_candidate: 448
## [1] 1471 448
## [1] 448 13
## Iteration: 93 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 93 cells_candidate: 1454 muts_candidate: 449
## [1] 1454 449
## [1] 449 13
## Iteration: 94 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 94 cells_candidate: 1444 muts_candidate: 449
## [1] 1444 449
## [1] 449 13
## Iteration: 95 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 95 cells_candidate: 1451 muts_candidate: 450
## [1] 1451 450
## [1] 450 13
## Iteration: 96 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 96 cells_candidate: 1459 muts_candidate: 454
## [1] 1459 454
## [1] 454 13
## Iteration: 97 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 97 cells_candidate: 1425 muts_candidate: 448
## [1] 1425 448
## [1] 448 13
## Iteration: 98 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 98 cells_candidate: 1438 muts_candidate: 457
## [1] 1438 457
## [1] 457 13
## Iteration: 99 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 99 cells_candidate: 1470 muts_candidate: 453
## [1] 1470 453
## [1] 453 13
## Iteration: 100 loss_cell_indices: 947 loss_mut_indices: 239
## Iteration: 100 cells_candidate: 1485 muts_candidate: 442
## [1] 1485 442
## [1] 442 13

# 3. merge res
data_scMutModel_res <- do.call(rbind, data_scMutModel_list)

# 4. column
data_scMutModel_res <- data_scMutModel_res %>%
  left_join(data_check_Meta, by = "Cellbarcode")

data_scMutModel_res <- data_scMutModel_res %>% dplyr::filter(Group %in% c("Prog", "prog_my", "prog_ery"))

table(data_check_Meta$Group)
##

```

```

##      ery1      ery2      ery3      ery4      ery5      ery6      mix      my1
##      59       83       61       67       63       94       37      161
##      my2      my3      my4      prog prog_ery  prog_my
##      365      429      308      122      18       26
# keep 100 cells for each cell type

data_scMutModel_res$GroupQun <- as.character(data_scMutModel_res$Group)

data_scMutModel_res$GroupQun[grepl("ery1|ery2", data_scMutModel_res$Group)] <- "ery1_2"
data_scMutModel_res$GroupQun[grepl("ery3|ery4", data_scMutModel_res$Group)] <- "ery3_4"
data_scMutModel_res$GroupQun[grepl("ery5|ery6", data_scMutModel_res$Group)] <- "ery5_6"

data_scMutModel_res$Group <- factor(data_scMutModel_res$Group, levels = c("Prog", "prog", "prog_my", "m
data_scMutModel_res$GroupQun <- factor(data_scMutModel_res$GroupQun, levels = c("Prog", "prog", "prog_my
data_scMutModel_res$CellMutTag <- factor(data_scMutModel_res$CellMutTag, levels = c("D_8", "D_14", "D_20

# 5. plot
data_scMutModel_res_summary <- data_scMutModel_res %>%
  dplyr::group_by(iteration, GroupQun) %>%
  dplyr::summarise(mean_scMutScore = mean(scMutScore, na.rm = TRUE), mean_EpiTraceAge = mean(EpiTraceAge
#dplyr::summarise(mean_EpiTraceAge = mean(EpiTraceAge_iterative, na.rm = TRUE)) %>%
  dplyr::ungroup()
## `summarise()` has grouped output by 'iteration'. You can override using the
## `.groups` argument.

data_scMutModel_res_summary <- data_scMutModel_res_summary %>%
  dplyr::mutate(
    mean_scMutScore_scaled = scales::rescale(mean_scMutScore, to = c(0, 1)),
    mean_EpiTraceAge_scaled = scales::rescale(mean_EpiTraceAge, to = c(0, 1))
  )

data_scMutModel_res_summary_Day <- data_scMutModel_res %>%
  dplyr::group_by(iteration, CellMutTag) %>%
  dplyr::summarise(mean_scMutScore = mean(scMutScore, na.rm = TRUE)) %>%
  dplyr::ungroup()
## `summarise()` has grouped output by 'iteration'. You can override using the
## `.groups` argument.

data_scMutModel_res_summary_Day <- data_scMutModel_res_summary_Day %>%
  dplyr::mutate(
    mean_scMutScore_scaled = scales::rescale(mean_scMutScore, to = c(0, 1))
  )

data_stability <- data_scMutModel_res %>%
  dplyr::group_by(GroupQun) %>%
  dplyr::summarise(
    mean_score = mean(scMutScore, na.rm = TRUE),
    sd_score = sd(scMutScore, na.rm = TRUE),
    cv_score = sd_score / mean_score
  )

p_scMutrace_iter_GroupQun <- ggplot(data = data_scMutModel_res_summary, mapping = aes(x = GroupQun, y =
  #geom_violin(trim = TRUE, alpha = 0.6) +

```

```

geom_boxplot(outliers = FALSE) +
ylim(0.4, 0.9) +
labs(x = "", y = "scMutraceScore") +
scale_fill_manual(values = cluster_color) +
theme_classic()

p_scMutrace_iter_GroupQunScaled <- ggplot(data = data_scMutModel_res_summary, mapping = aes(x = GroupQun, y = scMutraceScore)) +
  #geom_violin(trim = TRUE, alpha = 0.6) +
  geom_boxplot(outliers = FALSE) +
  ylim(0, 1) +
  labs(x = "", y = "mean_scMutScore_scaled") +
  scale_fill_manual(values = cluster_color) +
  theme_classic()

p_scMutrace_iter_Day <- ggplot(data = data_scMutModel_res_summary_Day, mapping = aes(x = CellMutTag, y = scMutraceScore)) +
  #geom_violin(trim = TRUE, alpha = 0.6) +
  geom_boxplot(outliers = FALSE) +
  ylim(0.4, 0.9) +
  labs(x = "", y = "scMutraceScore") +
  scale_fill_manual(values = day_color) +
  theme_classic()

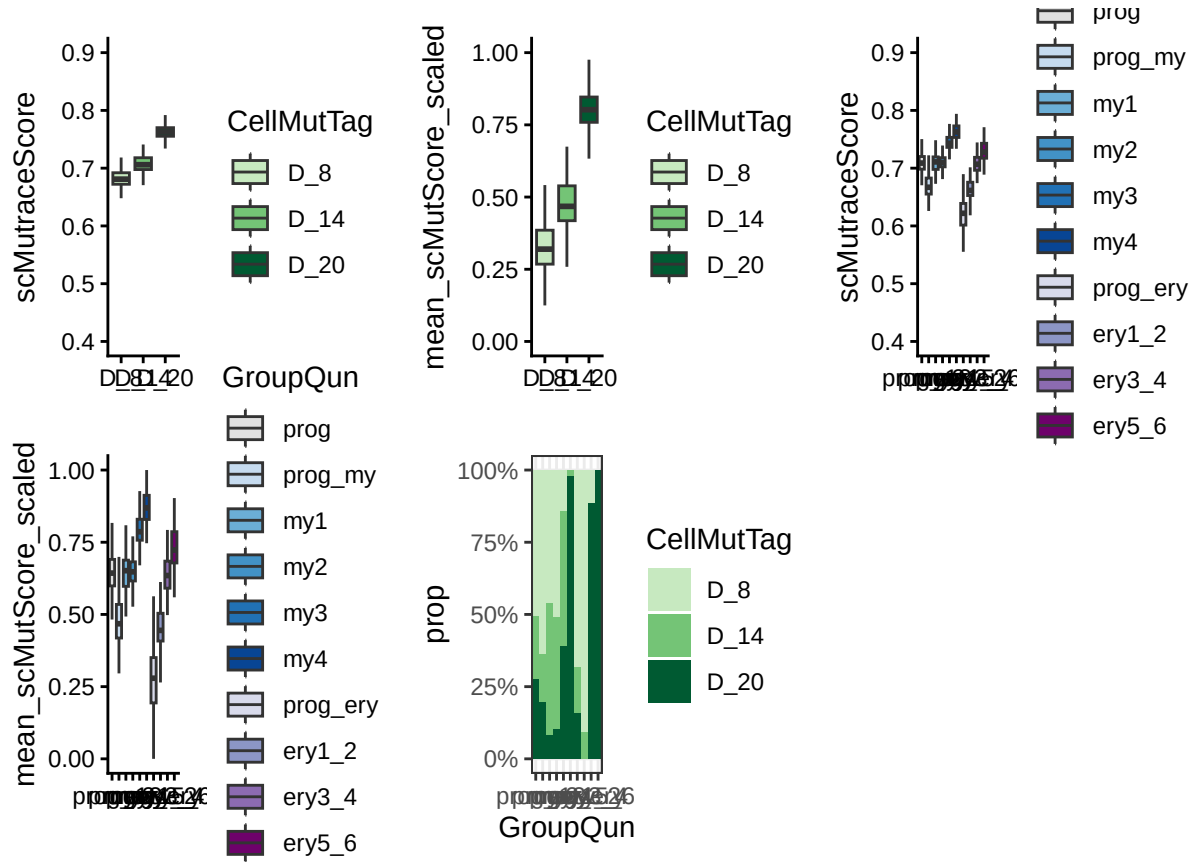
p_scMutrace_iter_Day_scaled <- ggplot(data = data_scMutModel_res_summary_Day, mapping = aes(x = CellMutTag, y = mean_scMutScore_scaled)) +
  #geom_violin(trim = TRUE, alpha = 0.6) +
  geom_boxplot(outliers = FALSE) +
  ylim(0, 1) +
  labs(x = "", y = "mean_scMutScore_scaled") +
  scale_fill_manual(values = day_color) +
  theme_classic()

df_iter_frac <- data_scMutModel_res %>%
  group_by(GroupQun, CellMutTag) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(GroupQun) %>%
  mutate(prop = n / sum(n))

p_iter_cellfrac <- ggplot(df_iter_frac, aes(x = GroupQun, y = prop, fill = CellMutTag)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_fill_manual(values = day_color) +
  scale_y_continuous(labels = scales::percent) +
  theme_bw()

p_scMutrace_iter_Day + p_scMutrace_iter_Day_scaled + p_scMutrace_iter_GroupQun + p_scMutrace_iter_GroupQun_scaled

```



10 randomMutNum

```
sample_size <- 100
iter_num <- 1:100
data_random_mutNum <- data.frame(iteration_num = iter_num, CD34_800_Day08 = 0,
                                CD34_800_Day14 = 0, CD34_800_Day20 = 0)

CD34_Meta_selectedRandom <- CD34_Meta[CD34_Meta$Cellbarcode %in% rownames(CD34_CSM),]

for (iter in iter_num){
  cell_08 <- CD34_Meta_selectedRandom %>% filter(CellMutTag == "D_8") %>% pull(Cellbarcode) %>% sample(
  cell_14 <- CD34_Meta_selectedRandom %>% filter(CellMutTag == "D_14") %>% pull(Cellbarcode) %>% sample(
  cell_20 <- CD34_Meta_selectedRandom %>% filter(CellMutTag == "D_20") %>% pull(Cellbarcode) %>% sample(
  mut_08 <- retrieveLocsByCells(CD34_CSM, cell_08)$Mut
  mut_14 <- retrieveLocsByCells(CD34_CSM, cell_14)$Mut
  mut_20 <- retrieveLocsByCells(CD34_CSM, cell_20)$Mut
  data_random_mutNum[iter, "CD34_800_Day08"] <- length(mut_08)
  data_random_mutNum[iter, "CD34_800_Day14"] <- length(mut_14)
  data_random_mutNum[iter, "CD34_800_Day20"] <- length(mut_20)
}

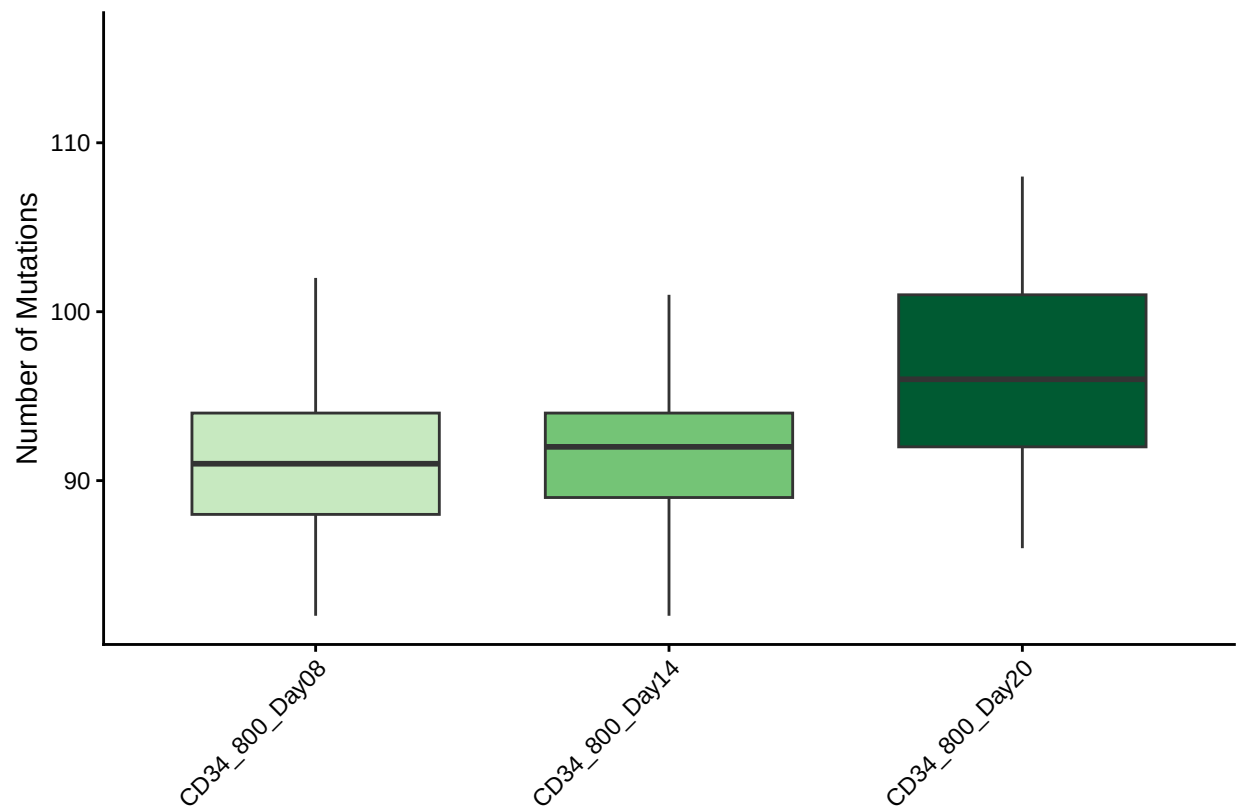
data_random_mutNum_long <- data_random_mutNum %>%
  pivot_longer(
```

```

cols = -iteration_num,
names_to = "Sample",
values_to = "Mutation_Number"
)

p_randomMutNum_est <- ggplot(data_random_mutNum_long, aes(x = Sample, y = Mutation_Number, fill = Sample)) +
  geom_boxplot(outlier.shape = NA, width = 0.7) +
  scale_fill_manual(values = c(CD34_800_Day08 = "#c7e9c0", CD34_800_Day14 = "#74c476", CD34_800_Day20 = "#1f77b4")) +
  #geom_jitter(width = 0.15, size = 1.5, alpha = 0.6) +
  theme_classic() +
  labs(
    x = "",
    y = "Number of Mutations"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )
p_randomMutNum_est

```



11 outputFigures

```
res_index <- res_index[res_index$Diversity > 0, ]

res_index %>% dplyr::filter(Group %in% c("Prog", "prog_my", "prog_ery", "prog", "my1", "my2", "my3", "my4"))
## `summarise()` has grouped output by 'clone', 'Group'. You can override using
## the `.groups` argument.
res_index_meta_summary_scMutScore$CultureDay <- paste("D_", res_index_meta_summary_scMutScore$cell_day, sep = "")

res_index_meta_summary_scMutScore$cell_group <- 'Prog'
res_index_meta_summary_scMutScore$cell_group[grepl('my1', res_index_meta_summary_scMutScore$Group)] <- '1'
res_index_meta_summary_scMutScore$cell_group[grepl('my2|my3|my4', res_index_meta_summary_scMutScore$Group)] <- '2'
res_index_meta_summary_scMutScore$cell_group <- factor(res_index_meta_summary_scMutScore$cell_group, levels = c('1', '2'))

res_index$GroupQun <- as.character(res_index$Group)

res_index$GroupQun[grepl("ery1|ery2", res_index$Group)] <- "ery1_2"
res_index$GroupQun[grepl("ery3|ery4", res_index$Group)] <- "ery3_4"
res_index$GroupQun[grepl("ery5|ery6", res_index$Group)] <- "ery5_6"

res_index$Group <- factor(res_index$Group, levels = c("Prog", "prog", "prog_my", "my1", "my2", "my3", "my4"))
res_index$GroupQun <- factor(res_index$GroupQun, levels = c("Prog", "prog", "prog_my", "my1", "my2", "my3", "my4"))
res_index_meta_summary_scMutScore$CultureDay <- factor(res_index_meta_summary_scMutScore$CultureDay, levels = c("D_1", "D_2"))

CD34_Meta_selected <- CD34_Meta[CD34_Meta$Cellbarcode %in% res_index$Cellbarcode,]
CD34_Meta_selected$GroupQun <- CD34_Meta_selected$Group
CD34_Meta_selected$GroupQun[grepl("ery1|ery2", CD34_Meta_selected$Group)] <- "ery1_2"
CD34_Meta_selected$GroupQun[grepl("ery3|ery4", CD34_Meta_selected$Group)] <- "ery3_4"
CD34_Meta_selected$GroupQun[grepl("ery5|ery6", CD34_Meta_selected$Group)] <- "ery5_6"

UMAP_Day8 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 8),
    aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

UMAP_Day14 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 14),
    aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

UMAP_Day20 <- ggplot(data = CD34_Meta_selected) +
  geom_point(aes(x = x, y = y), color = "grey90", size = 1.5) +
  geom_point(data = subset(CD34_Meta_selected, cell_day == 20),
    aes(x = x, y = y, color = Group), size = 1.5, alpha = 0.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()
```

```

p_UMAP_Days <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = CellMutTag)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = day_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_Groups <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = Group)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_GroupsQun <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = GroupQun)) +
  geom_point(size = 1.5) +
  scale_color_manual(values = cluster_color) +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_pseudotime_my <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = pseudotime)) +
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option = "plasma") +
  xlab("") + ylab("") +
  theme_classic()

p_UMAP_pseudotime_ery <- ggplot(data = CD34_Meta_selected, mapping = aes(x = x, y = y, color = pseudotime)) +
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option = "plasma") +
  xlab("") + ylab("") +
  theme_classic()

# plot scMutScore by days
p_scMutScorebydays <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=scMutScore, fill = CellMutTag)) +
  scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

# plot scMutScore by cell types
p_scMutScorebycelltypes <- ggplot(res_index, aes(x=as.factor(Group), y=scMutScore, fill = Group)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_scMutScorebycelltypesQun <- ggplot(res_index, aes(x=as.factor(GroupQun), y=scMutScore, fill = GroupQun)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  ylim(0,1) +
  theme_classic()

p_EpiScorebyCelltype <- ggplot(res_index, aes(x=as.factor(Group), y=EpiTraceAge_iterative, fill = Group)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_EpiScorebyCelltypeQun <- ggplot(res_index, aes(x=as.factor(GroupQun), y=EpiTraceAge_iterative, fill = GroupQun)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

```

```

theme_classic()

p_CloneLevelByCellGroup <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = as.factor(cell_group), y = scMutScore)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "scMutScore") +
  theme_classic()

p_CloneLevelByDaysSplitbyCellgroup <- ggplot(res_index_meta_summary_scMutScore, aes(x=factor(cell_day), y=scMutScore)) +
  geom_line(aes(group=clone), size=0.5, alpha=0.1, linetype='dashed') + geom_boxplot(outlier.alpha = 1, fill="white") +
  facet_wrap(~cell_group) + theme_classic() + xlab('Days in culture') + ylab('Mean scMutScore') + theme(text = element_size(12))
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

p_CloneLevelByDaysSplitbyCellType <- ggplot(res_index_meta_summary_scMutScore, aes(x=factor(cell_day), y=scMutScore)) +
  geom_line(aes(group=clone), size=0.5, alpha=0.1, linetype='dashed') + geom_boxplot(outlier.alpha = 1, fill="white") +
  facet_wrap(~Group) + theme_classic() + xlab('Days in culture') + ylab('Mean scMutScore') + theme(text = element_size(12))

p_CloneLevelByDays <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = as.factor(CultureDay), y = scMutScore)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = day_color) +
  labs(x = "", y = "scMutScore") +
  theme_classic()

# EpiTraceAge
res_index %>% dplyr::group_by(clone, Group, cell_day) %>% dplyr::summarise(meanEpi=mean(EpiTraceAge_iterate))
## `summarise()` has grouped output by 'clone', 'Group'. You can override using
## the `.groups` argument.
res_index_meta_summary_EpiTraceAge$cell_group <- 'Prog'
res_index_meta_summary_EpiTraceAge$cell_group[grepl('my1', res_index_meta_summary_EpiTraceAge$Group)] <- 'my1'
res_index_meta_summary_EpiTraceAge$cell_group[grepl('my2|my3|my4', res_index_meta_summary_EpiTraceAge$Group)] <- 'my2|my3|my4'
res_index_meta_summary_EpiTraceAge$cell_group <- factor(res_index_meta_summary_EpiTraceAge$cell_group, levels=c('Prog', 'my1', 'my2|my3|my4'))

p_CloneLevelByCellGroup_EpiTraceAge <- ggplot(res_index_meta_summary_EpiTraceAge, mapping = aes(x = as.factor(clone), y = EpiTraceAge)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "EpiTraceAge") +
  theme_classic()

p_CloneLevelByCellGroup_scMuttraceScore <- ggplot(res_index_meta_summary_scMutScore, mapping = aes(x = as.factor(clone), y = scMutScore)) +
  geom_boxplot(outliers = TRUE) +
  geom_line(aes(group = clone), alpha = 0.1, linetype='dashed') +
  scale_fill_manual(values = cell_group_color) +
  labs(x = "", y = "EpiTraceAge") +
  theme_classic()

p_pseudotimeByDaysEry <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=pseudotime_ery, fill = CellMutTag)) +

```

```

    scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
    theme_classic()

p_pseudotimeByDaysMy <- ggplot(res_index, aes(x=as.factor(CellMutTag), y=pseudotime_my, fill = CellMutTag)) +
  scale_fill_manual(values = day_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_pseudotimeByGroupQunEry <- ggplot(res_index, aes(x=as.factor(GroupQun), y=pseudotime_ery, fill = GroupQun)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_pseudotimeByGroupQunMy <- ggplot(res_index, aes(x=as.factor(GroupQun), y=pseudotime_my, fill = GroupQun)) +
  scale_fill_manual(values = cluster_color) + geom_boxplot(outlier.alpha = 0) +
  theme_classic()

p_UMAP_scMutrace <- ggplot(data = res_index, mapping = aes(x = x, y = y, color = scMutScore)) +
  geom_point(size = 1.5) +
  #scale_color_manual(values = day_color) +
  scale_color_viridis(option="rocket") +
  xlab("") + ylab("") +
  theme_classic()

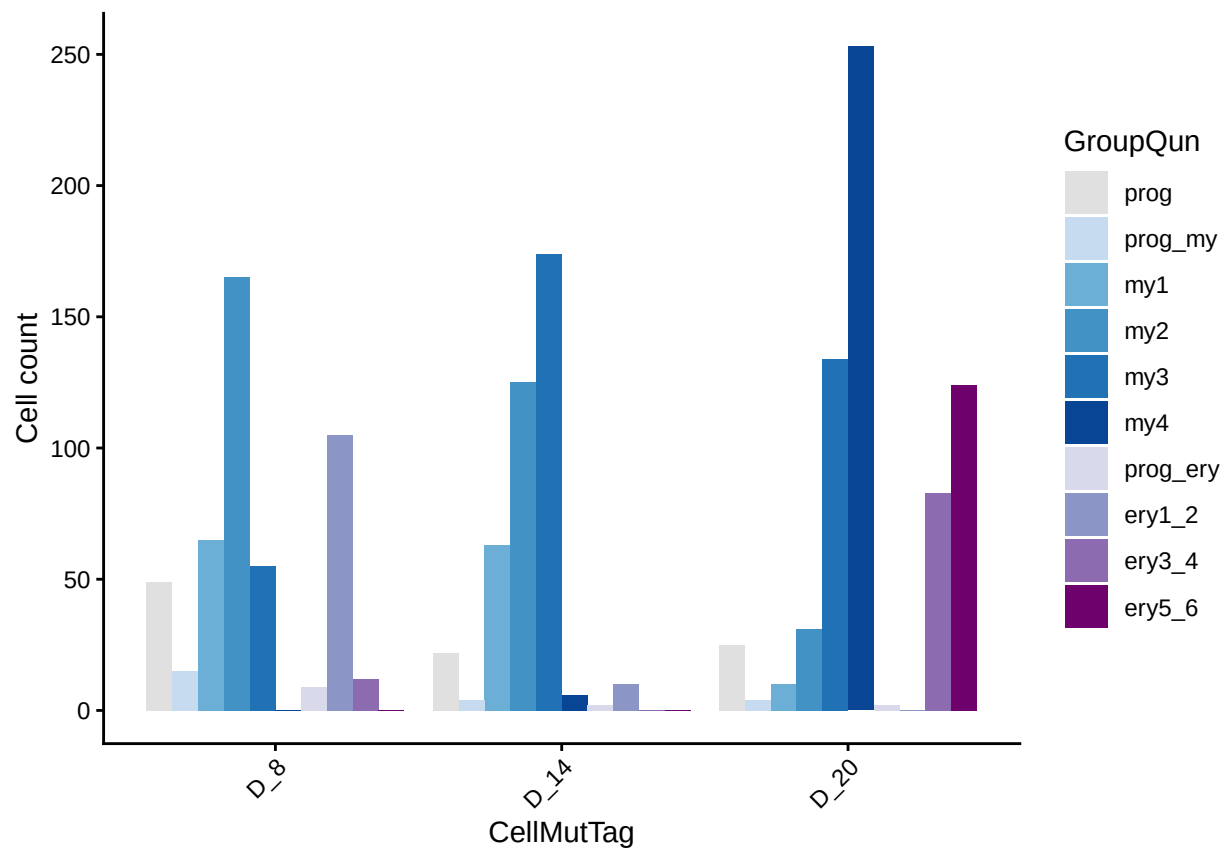
df_plot <- res_index %>%
  count(CellMutTag, GroupQun) %>%
  complete(
    CellMutTag,
    GroupQun = c(
      "prog", "prog_my",
      "my1", "my2", "my3", "my4",
      "prog_ery", "ery1_2", "ery3_4", "ery5_6"
    ),
    fill = list(n = 0)
  )

df_plot$GroupQun <- factor(
  df_plot$GroupQun,
  levels = c(
    "prog", "prog_my",
    "my1", "my2", "my3", "my4",
    "prog_ery", "ery1_2", "ery3_4", "ery5_6"
  )
)

ggplot(df_plot, aes(x = CellMutTag, y = n, fill = GroupQun)) +
  geom_col(position = "dodge") +
  labs(
    x = "CellMutTag",
    y = "Cell count",
    fill = "GroupQun"
  ) +
  scale_fill_manual(values = cluster_color) +
  theme_classic() +
  theme(

```

```
axis.text.x = element_text(angle = 45, hjust = 1)
)
```

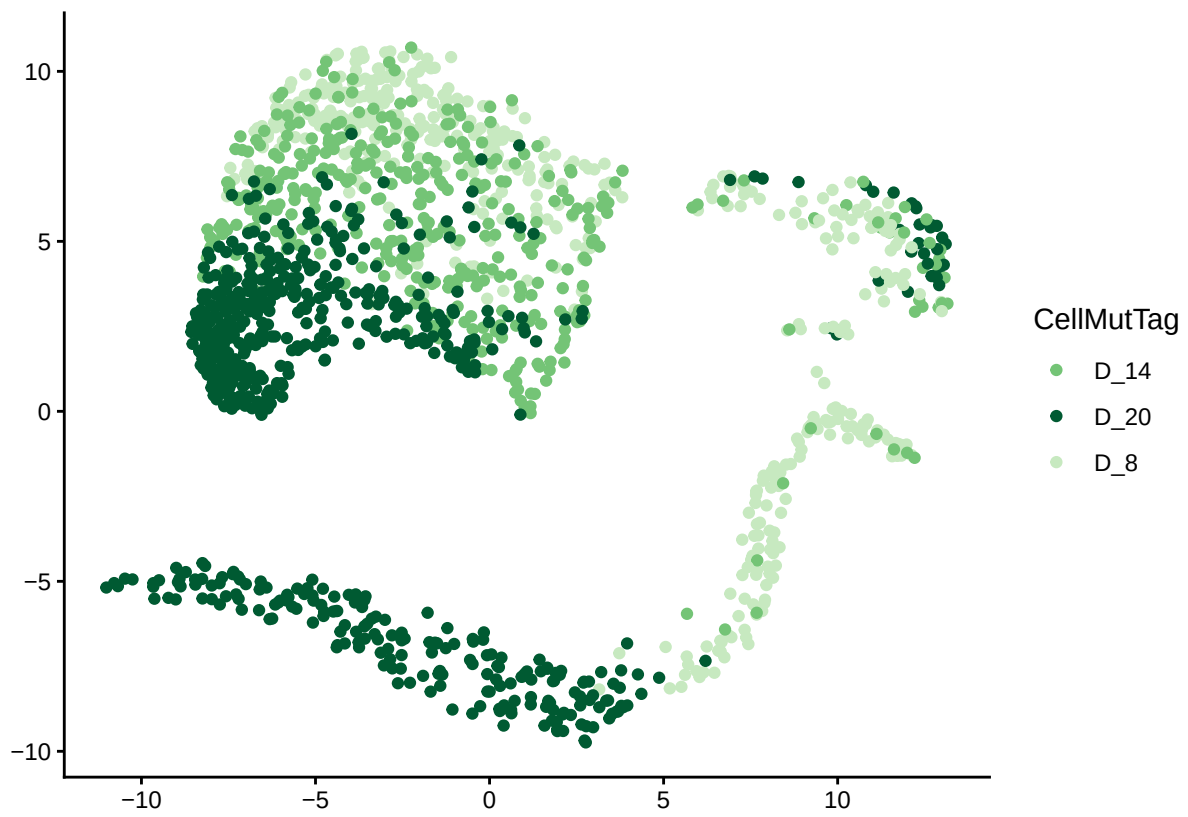


```
df_mean <- res_index %>%
  group_by(GroupQun) %>%
  summarise(
    mean_scMutScore = mean(scMutScore, na.rm = TRUE),
    mean_EpiTraceAge = mean(EpiTraceAge_iterative, na.rm = TRUE),
    mean_pseudotime_my = mean(pseudotime_my, na.rm = TRUE),
    mean_pseudotime_ery = mean(pseudotime_ery, na.rm = TRUE),
    n = n()
  )

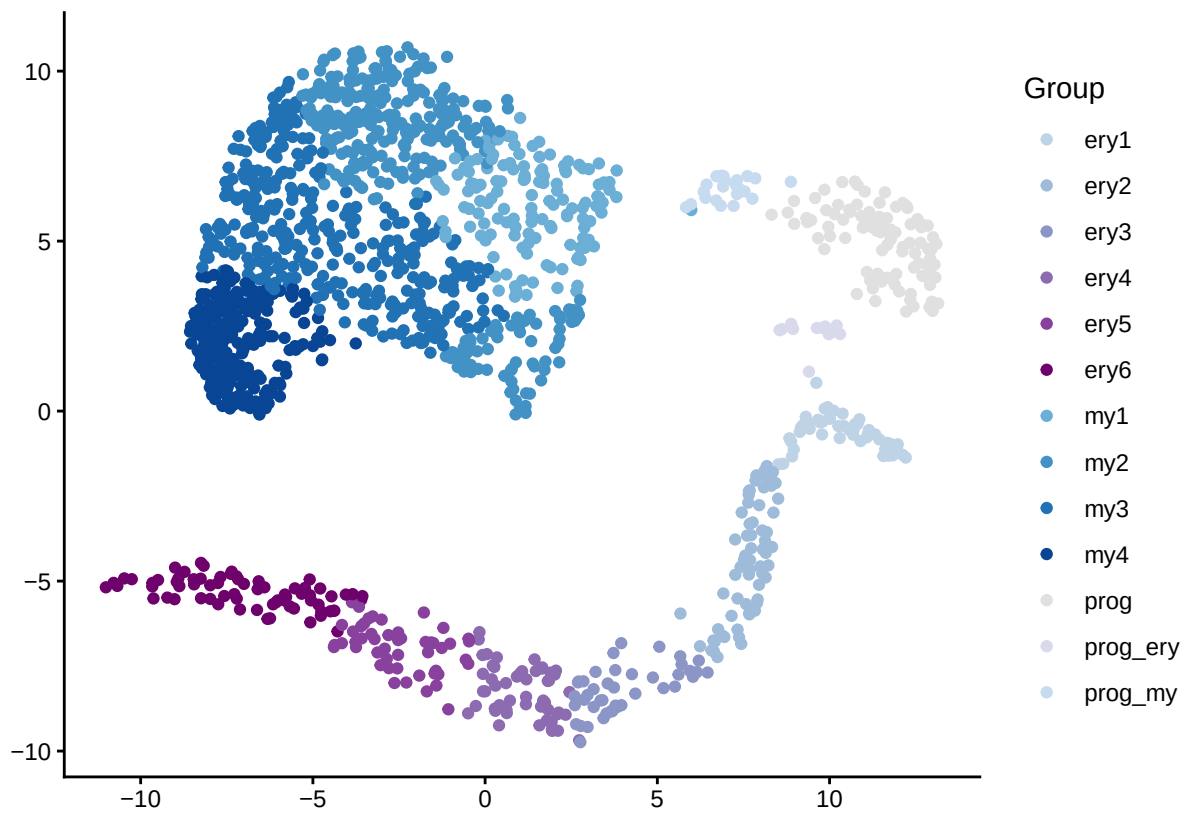
df_frac <- res_index %>%
  group_by(GroupQun, CellMutTag) %>%
  summarise(n = n(), .groups = "drop") %>%
  group_by(GroupQun) %>%
  mutate(prop = n / sum(n))

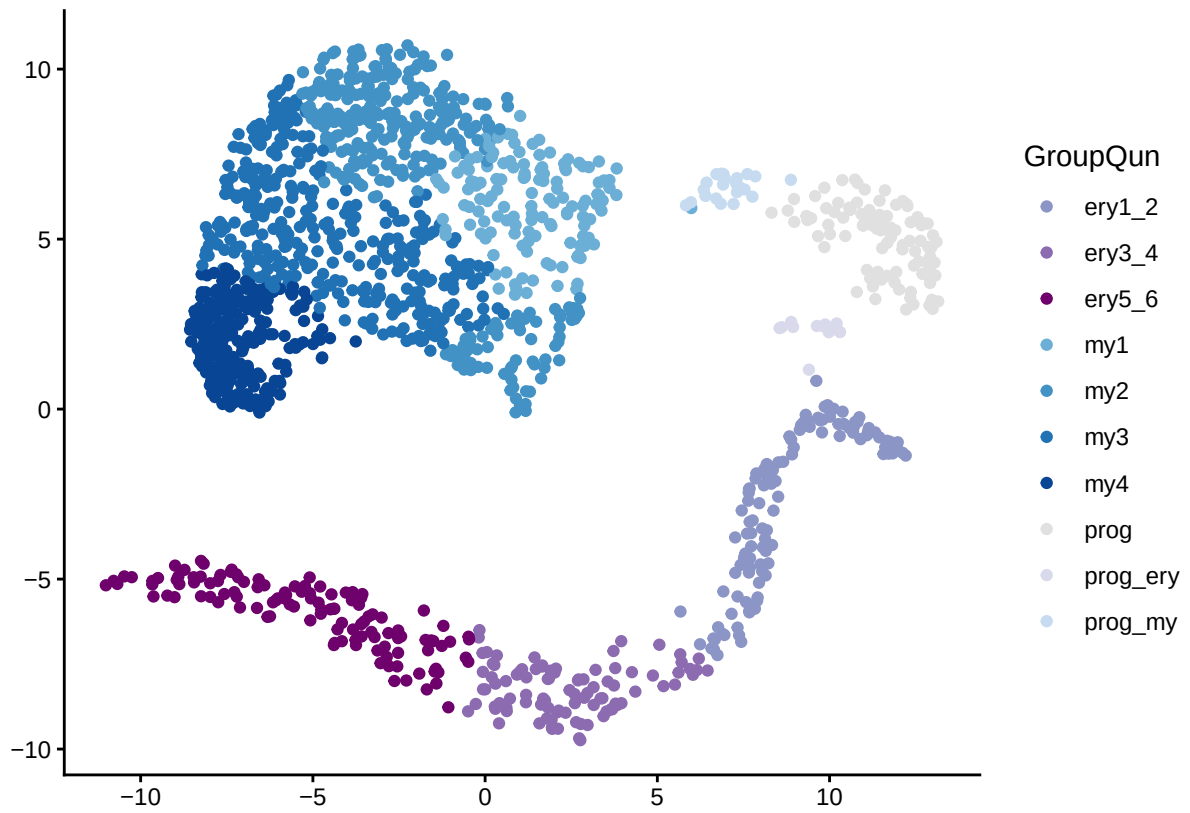
p_cellfrac <- ggplot(df_frac, aes(x = GroupQun, y = prop, fill = CellMutTag)) +
  geom_bar(stat = "identity", position = "fill") +
  scale_fill_manual(values = day_color) +
  scale_y_continuous(labels = scales::percent) +
  theme_bw()
```

p_UMAP_Days

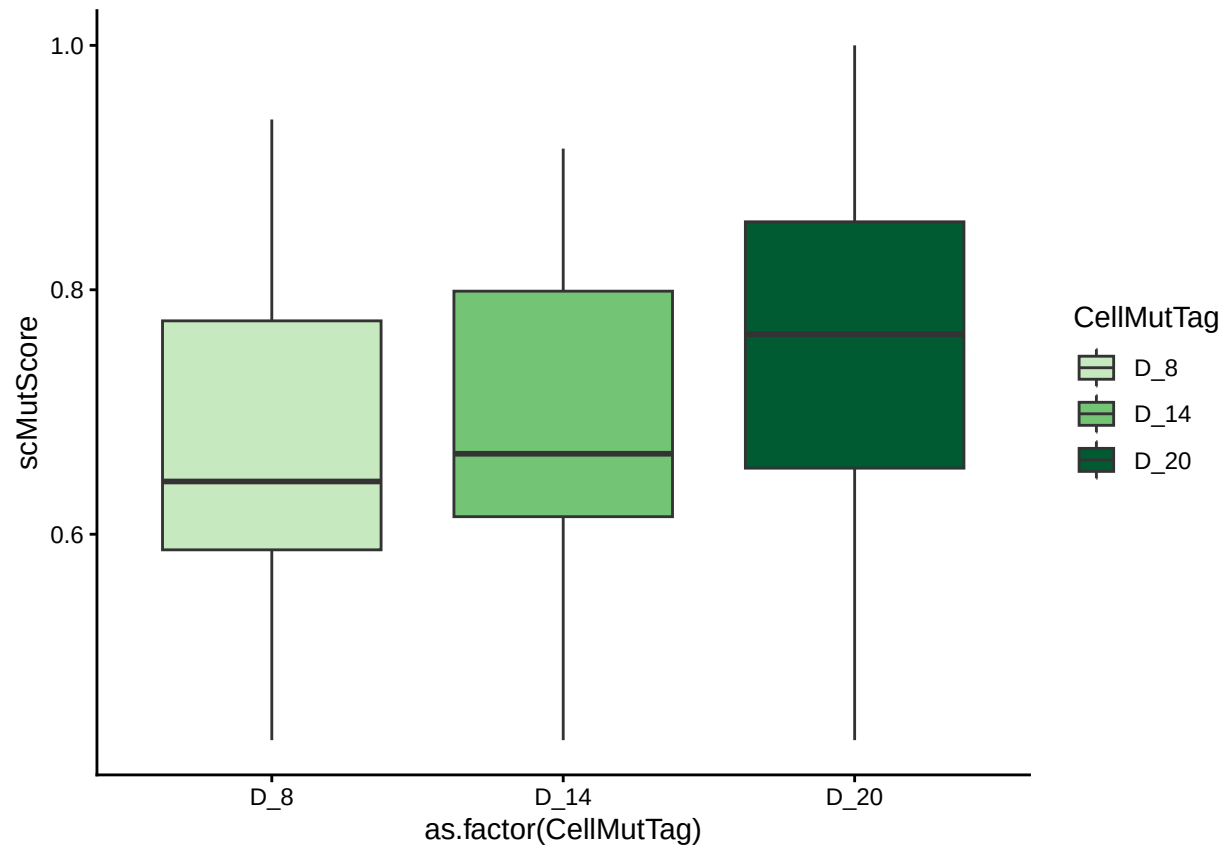


p_UMAP_Groups

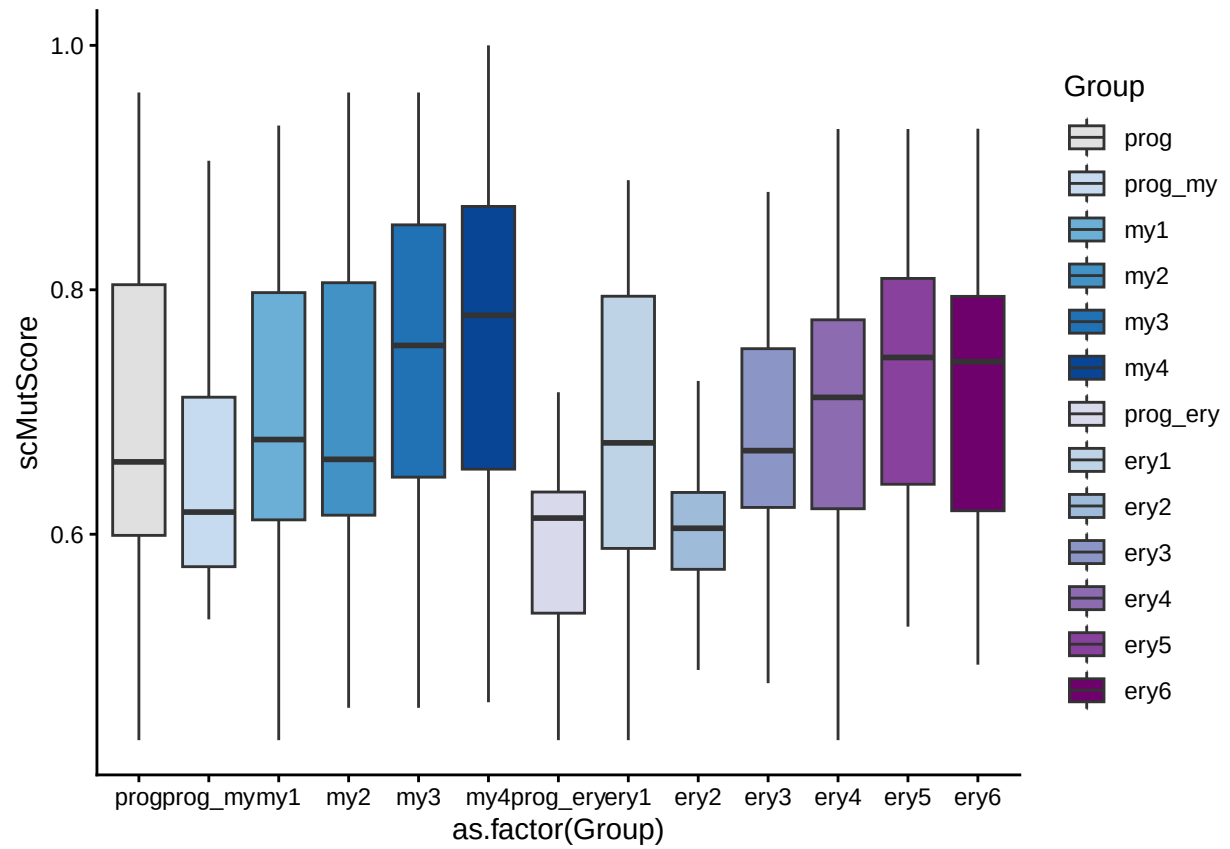




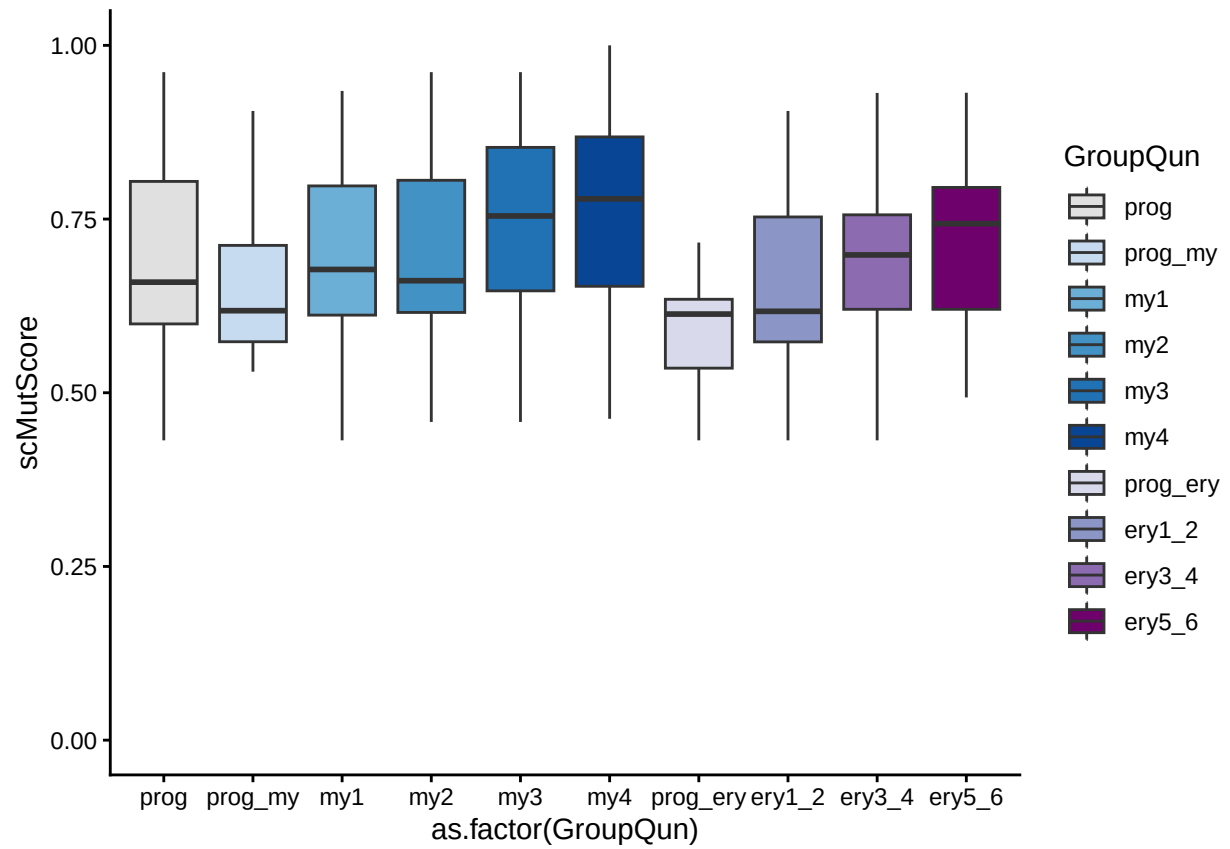
p_scMutScorebydays



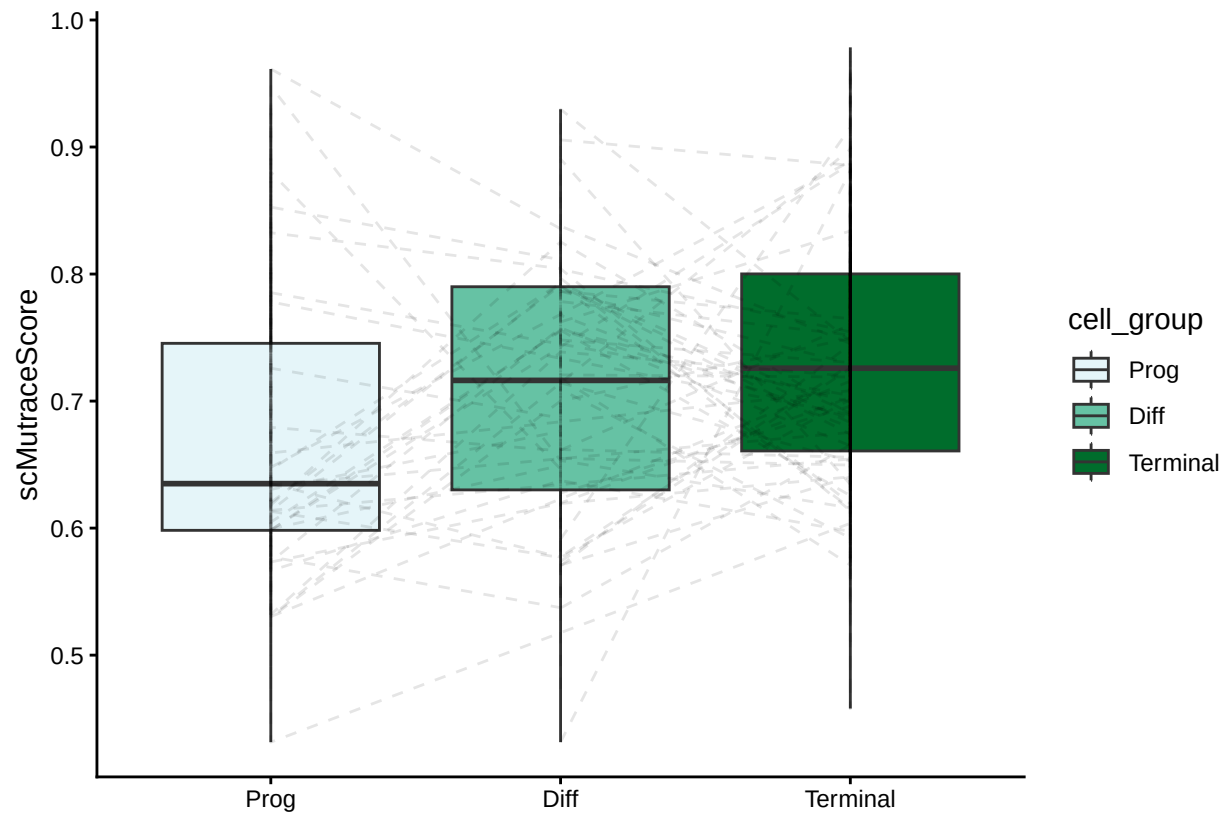
p_scMutScorebycelltypes



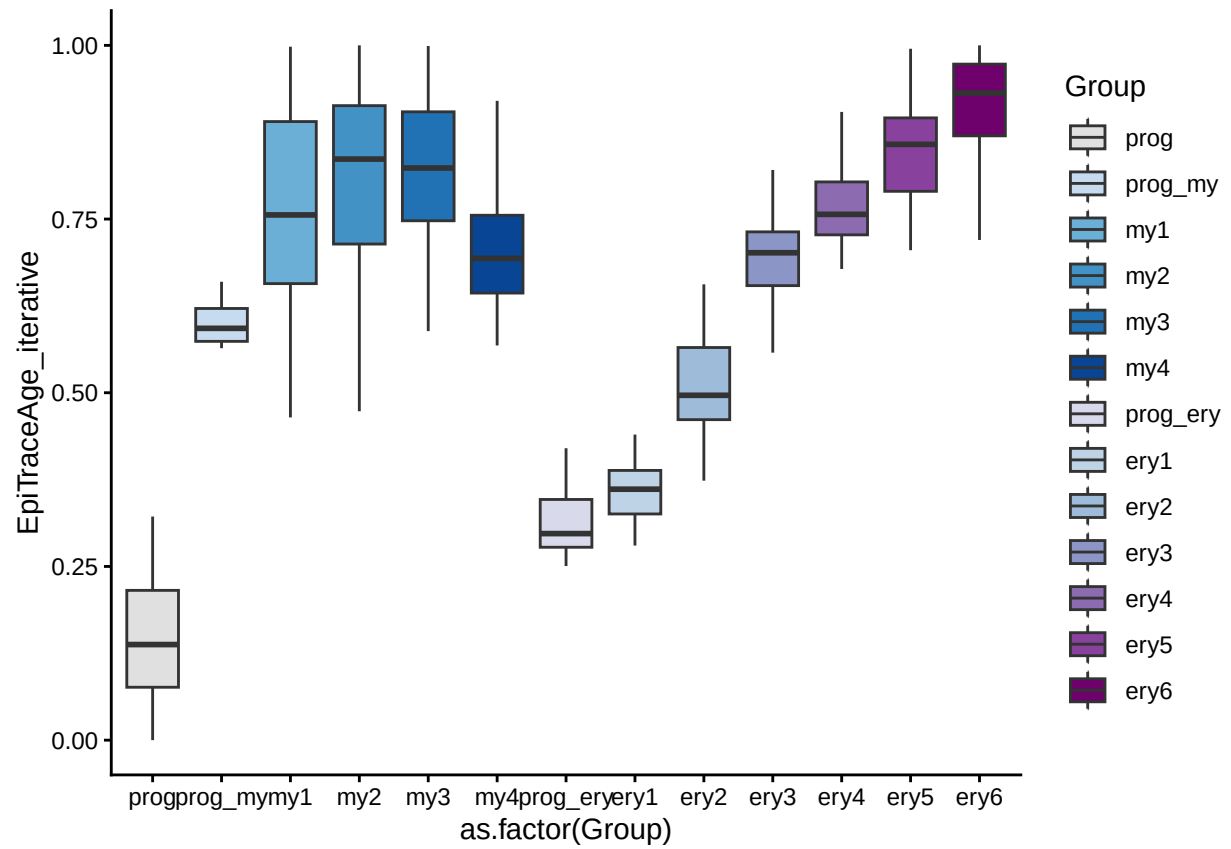
p_scMutScorebycelltypesQun



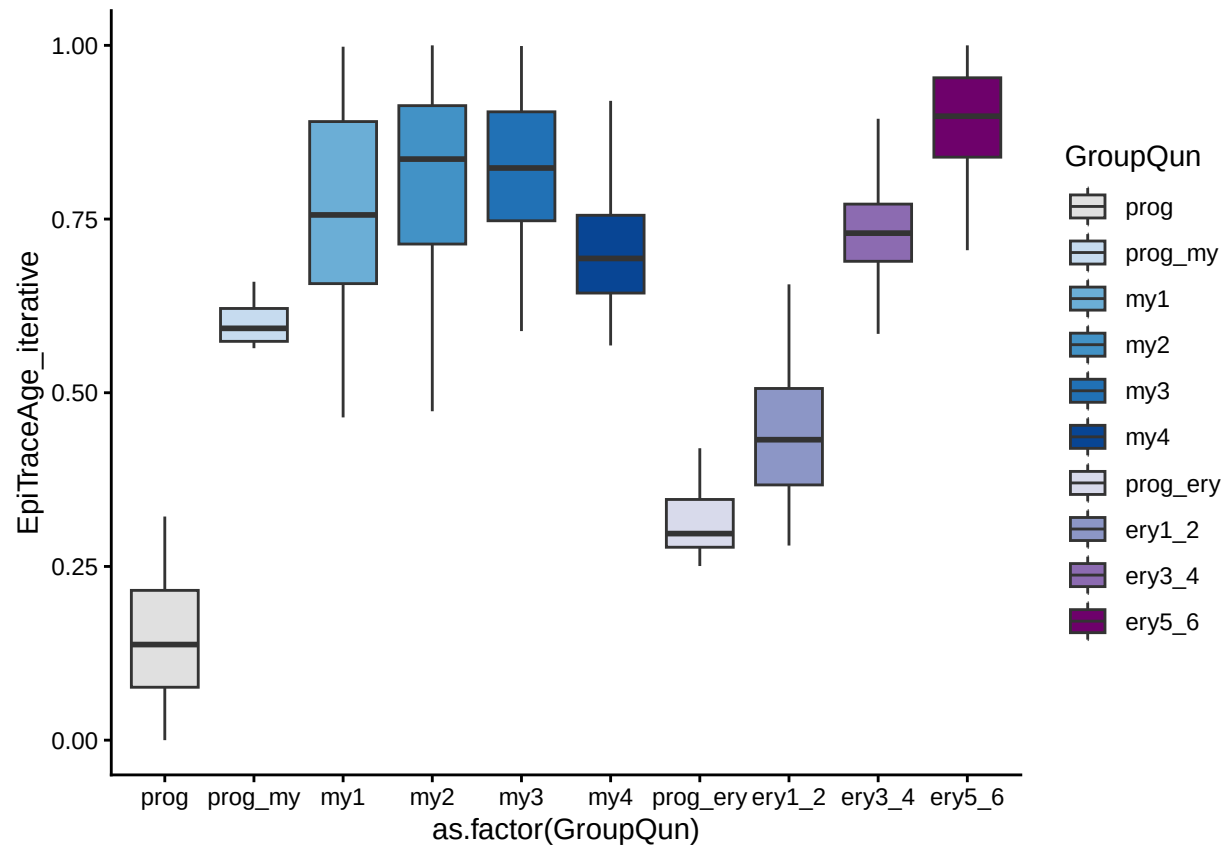
p_CloneLevelByCellGroup



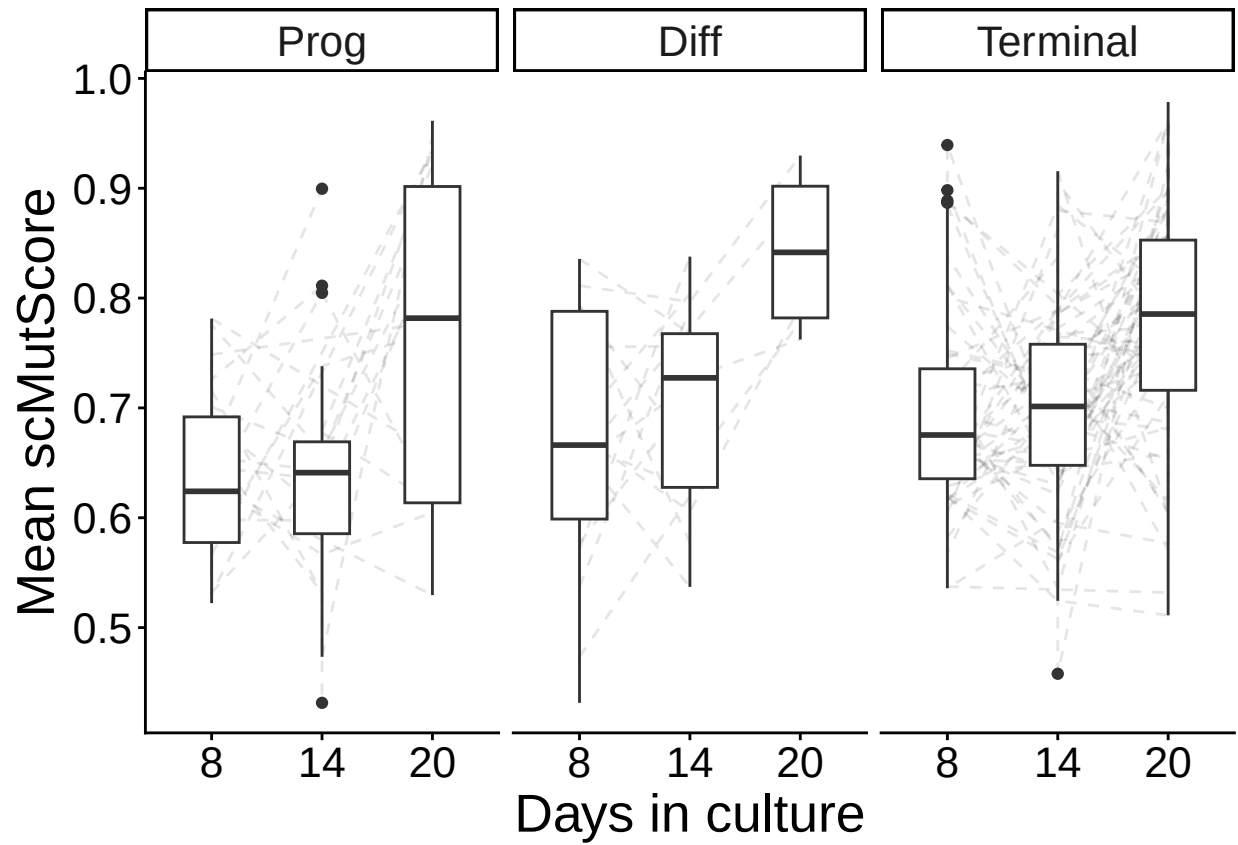
p_EpiScorebyCelltype



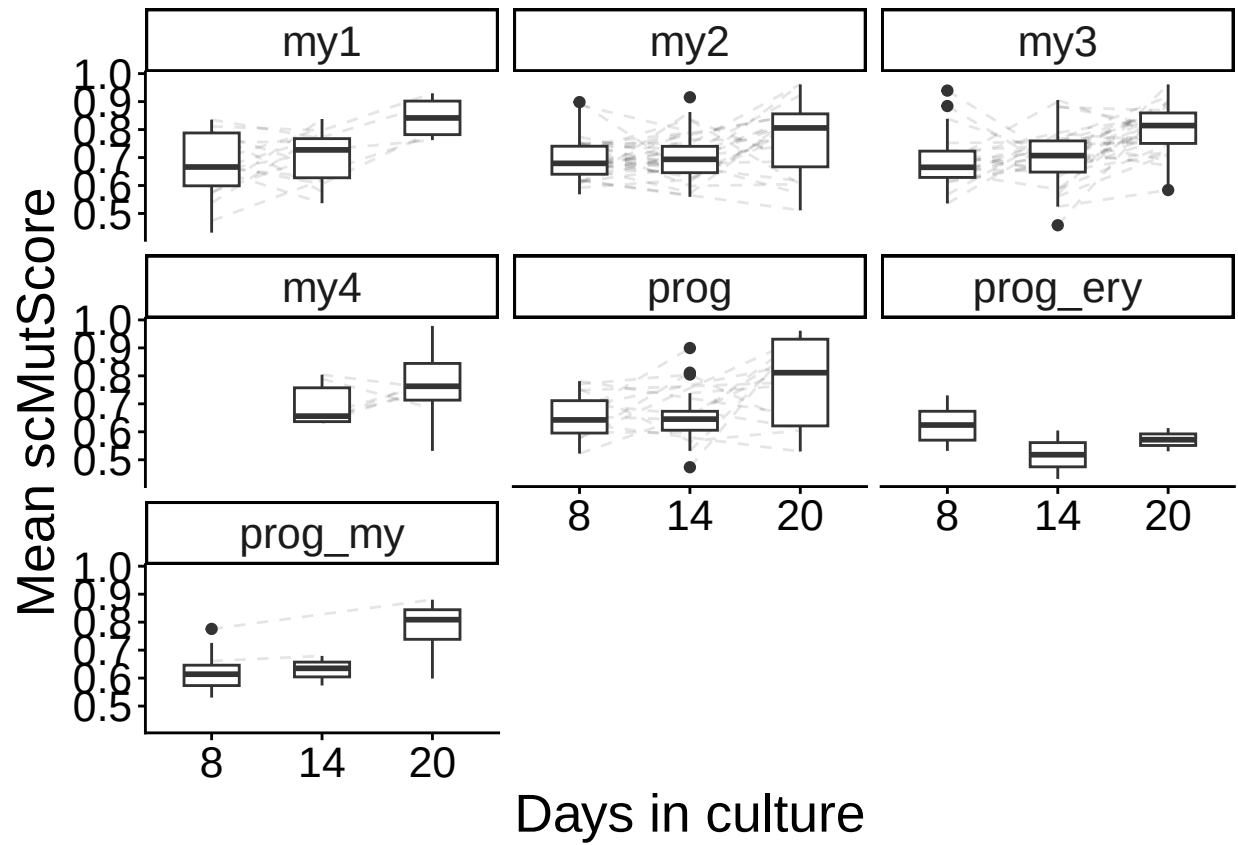
p_EpiScorebyCelltypeQun



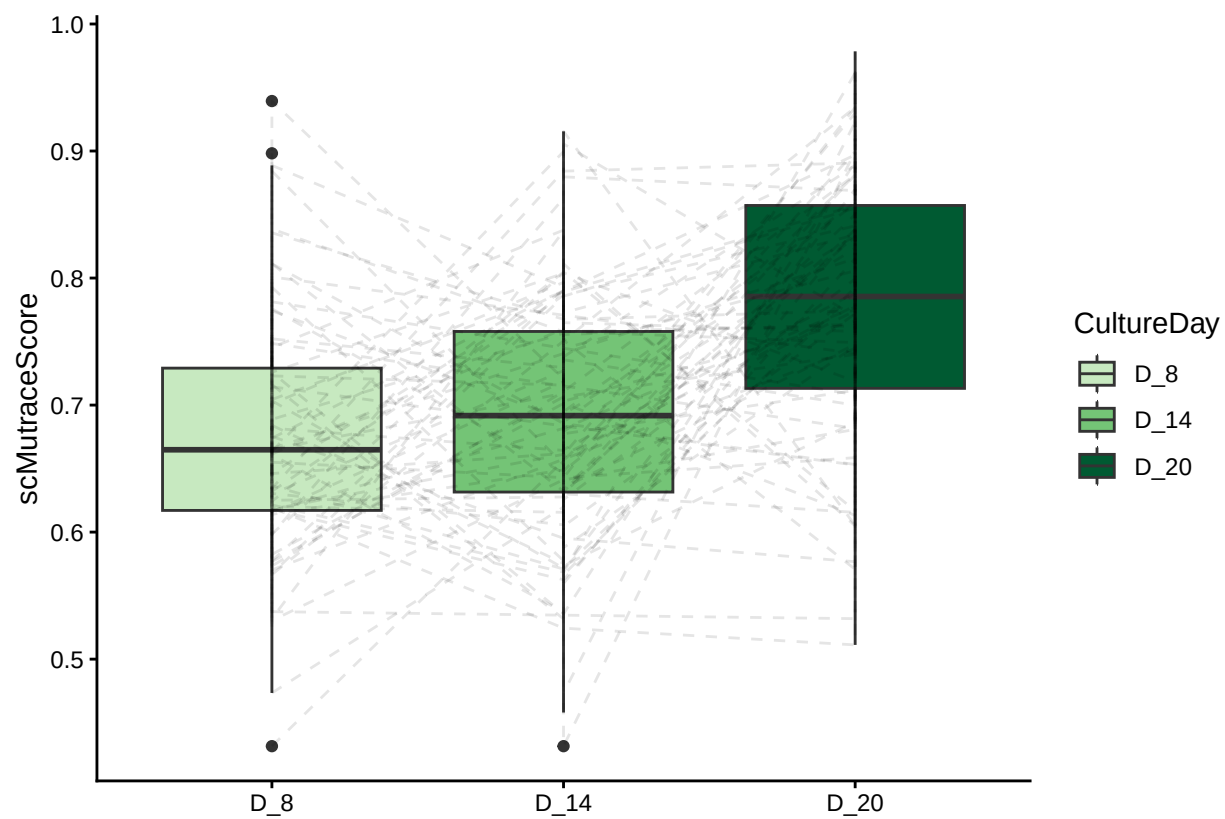
p_CloneLevelByDaysSplitbyCellgroup



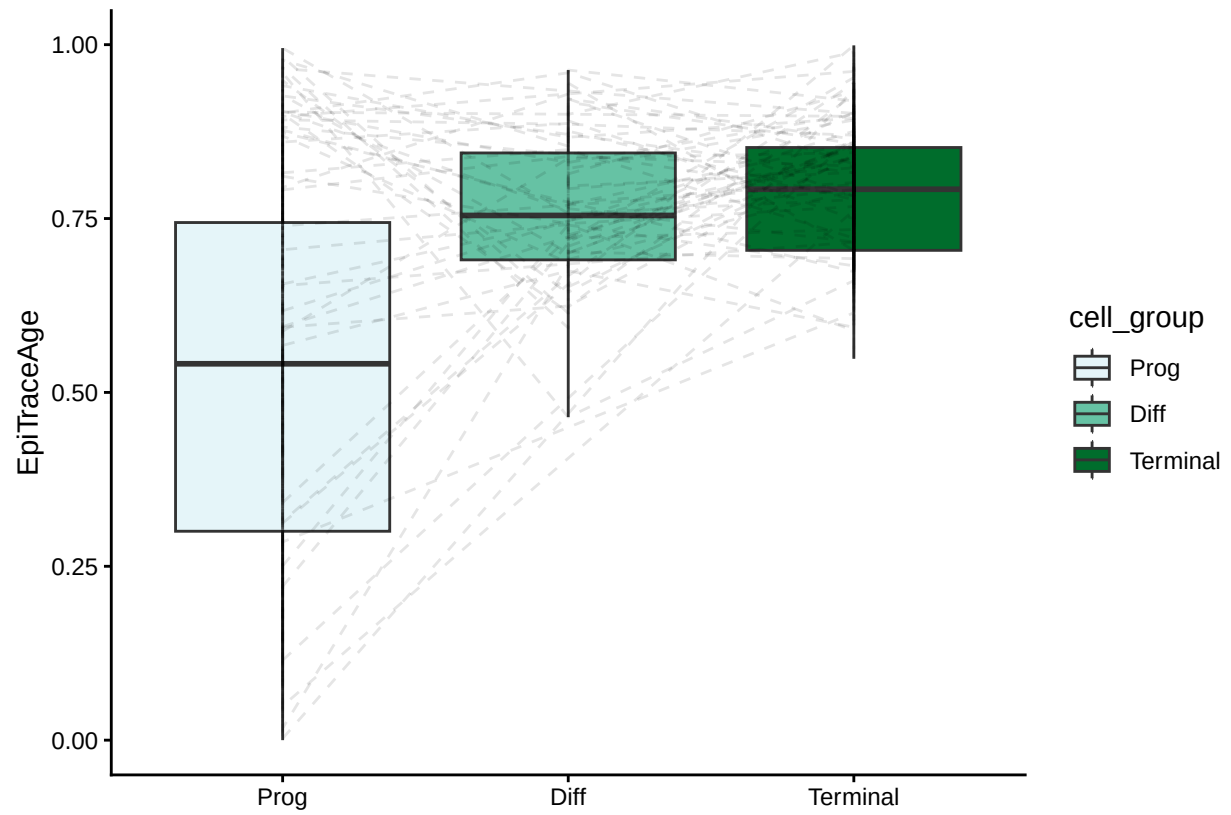
```
p_CloneLevelByDaysSplitbyCellType
## `geom_line()`: Each group consists of only one observation.
## i Do you need to adjust the group aesthetic?
```



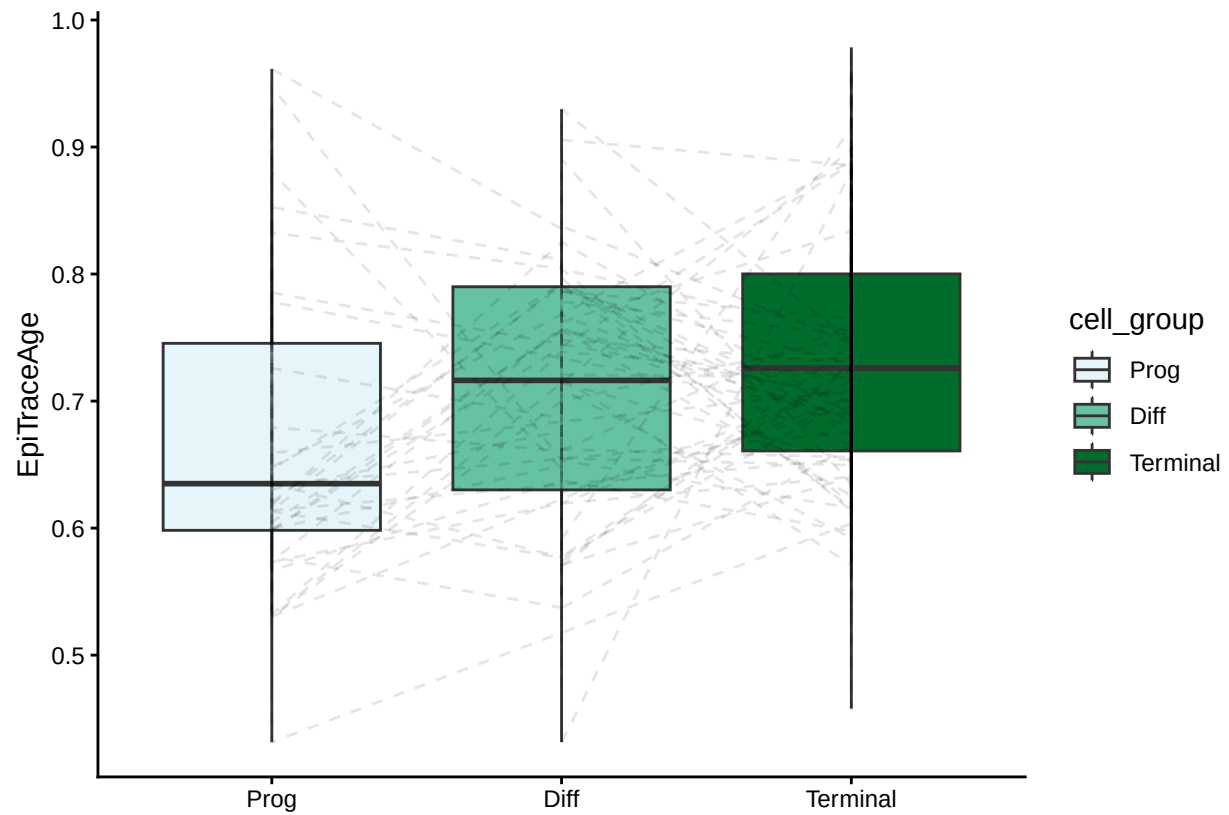
p_CloneLevelByDays



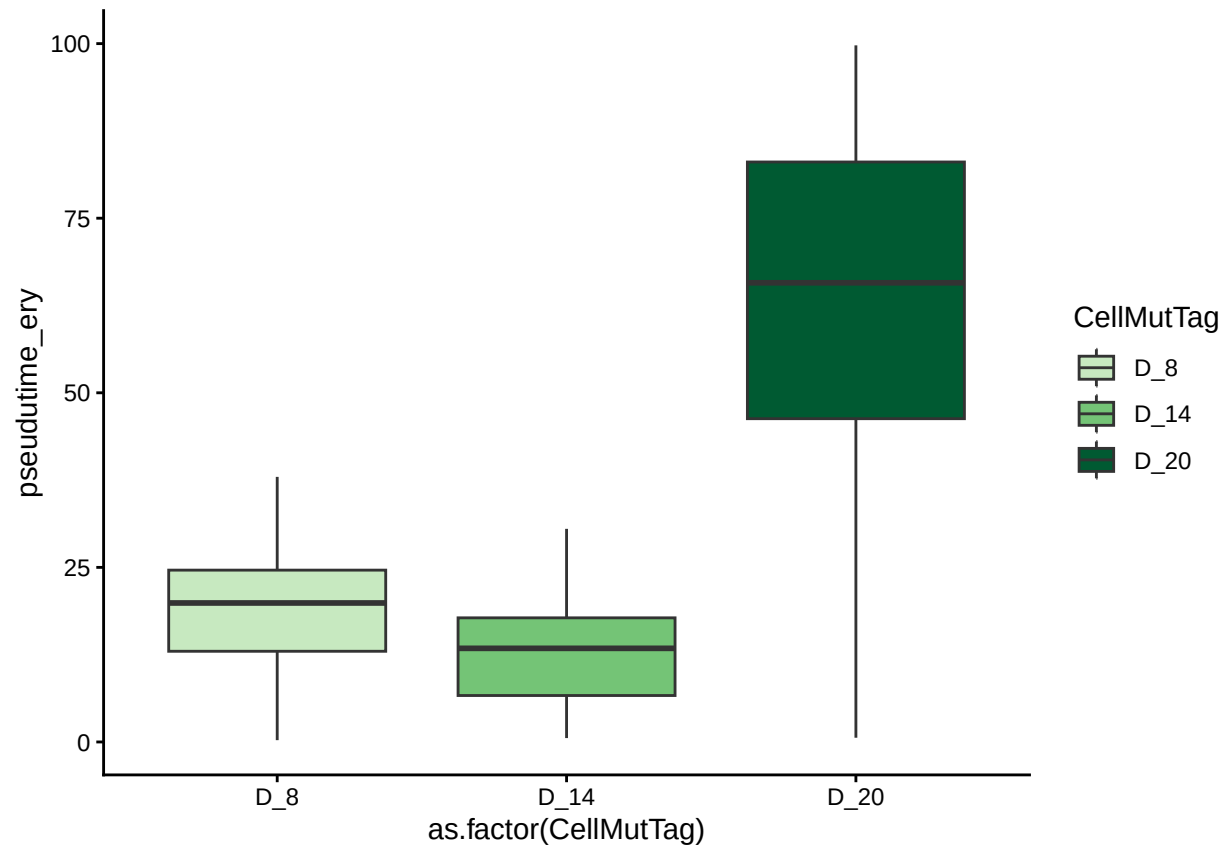
p_CloneLevelByCellGroup_EpiTraceAge



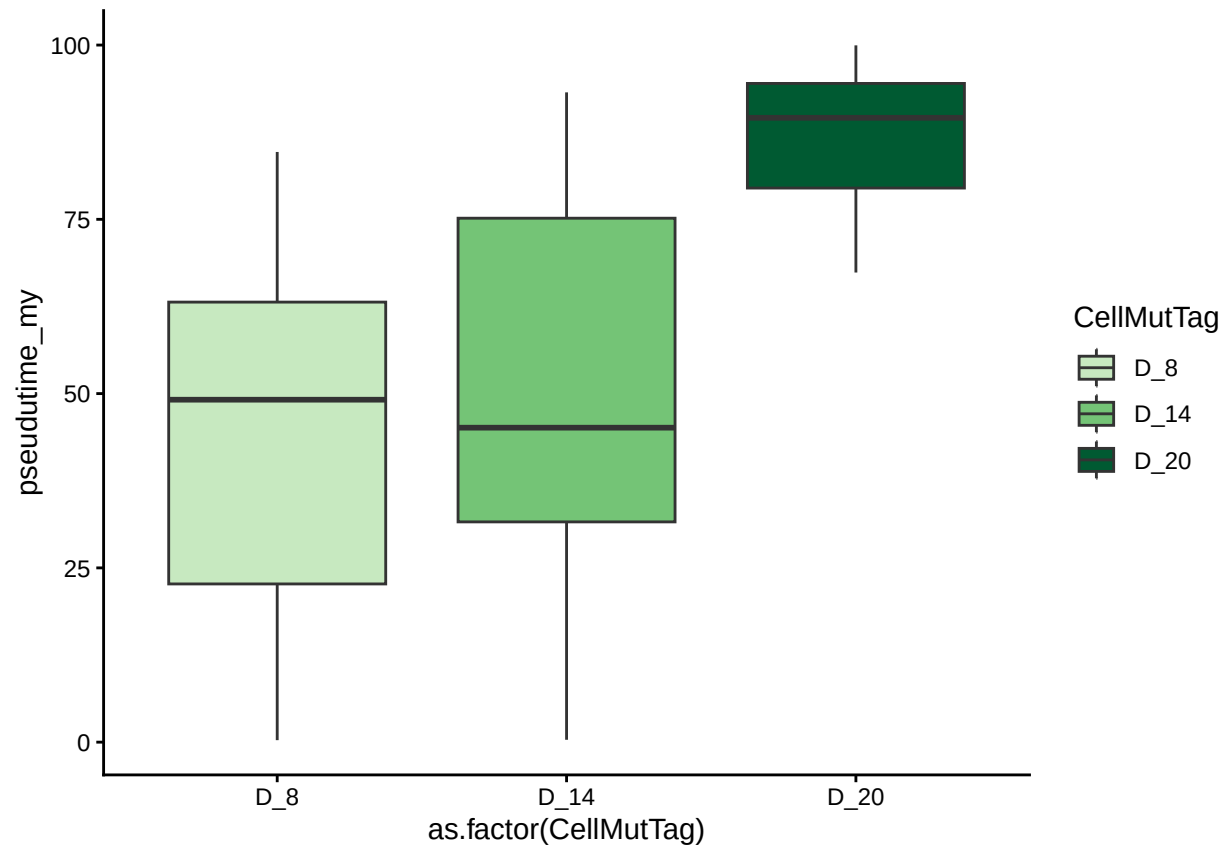
p_CloneLevelByCellGroup_scMutraceScore



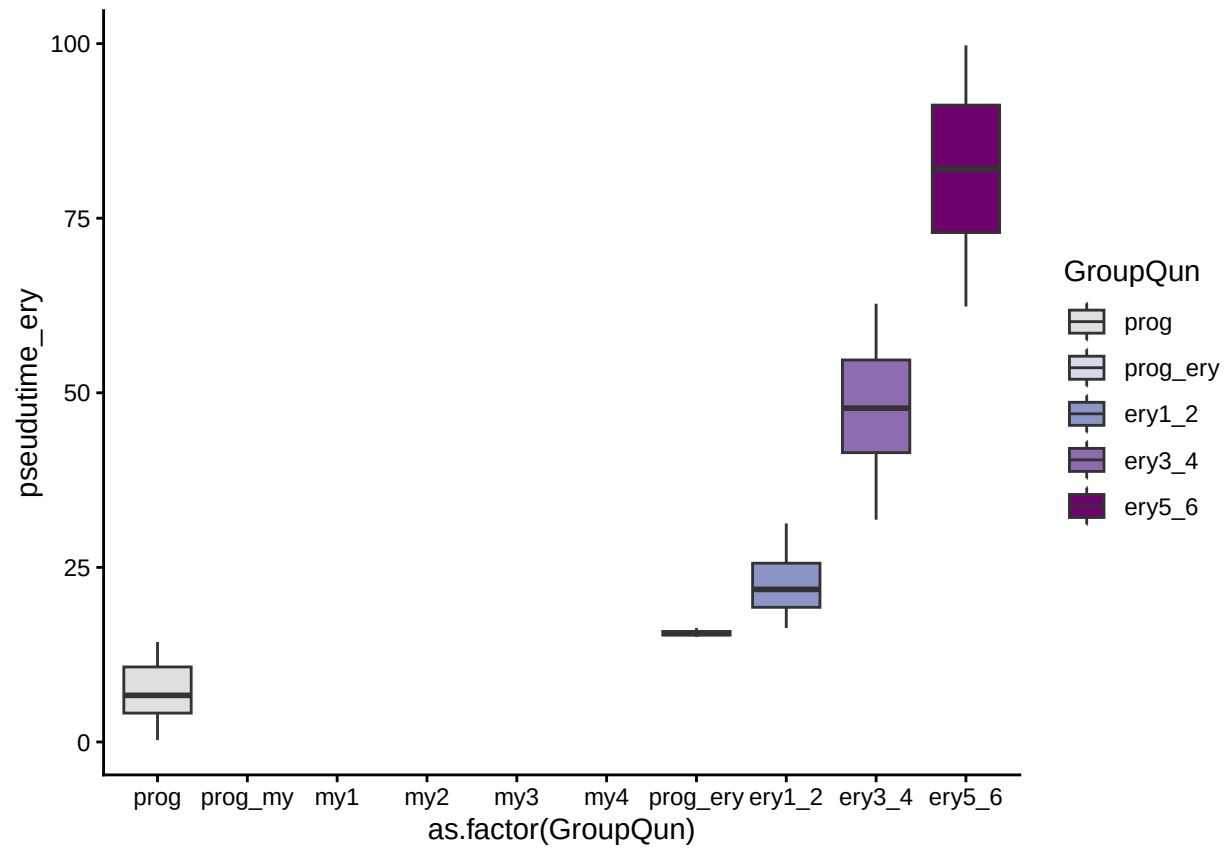
```
p_pseudotimeByDaysEry
## Warning: Removed 1104 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



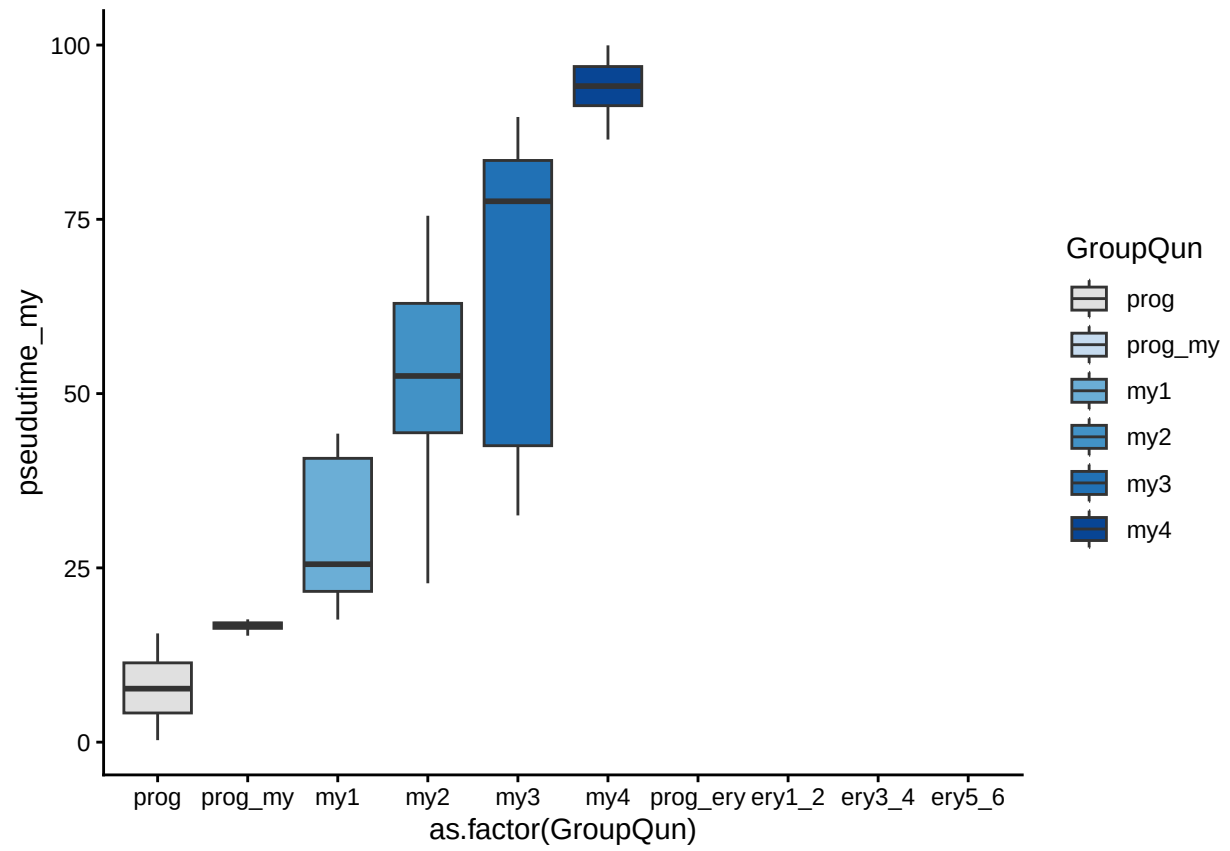
```
p_pseudotimeByDaysMy  
## Warning: Removed 347 rows containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



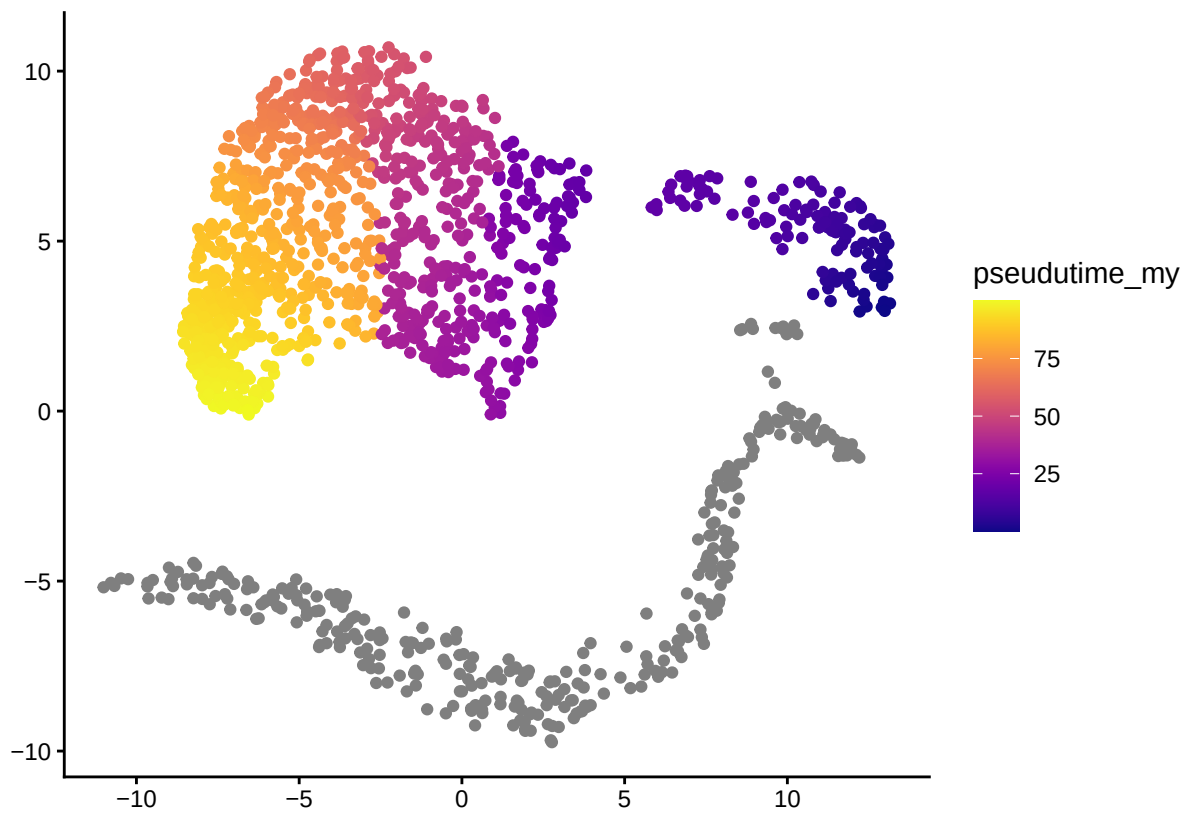
```
p_pseudotimeByGroupQunEry
## Warning: Removed 1104 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



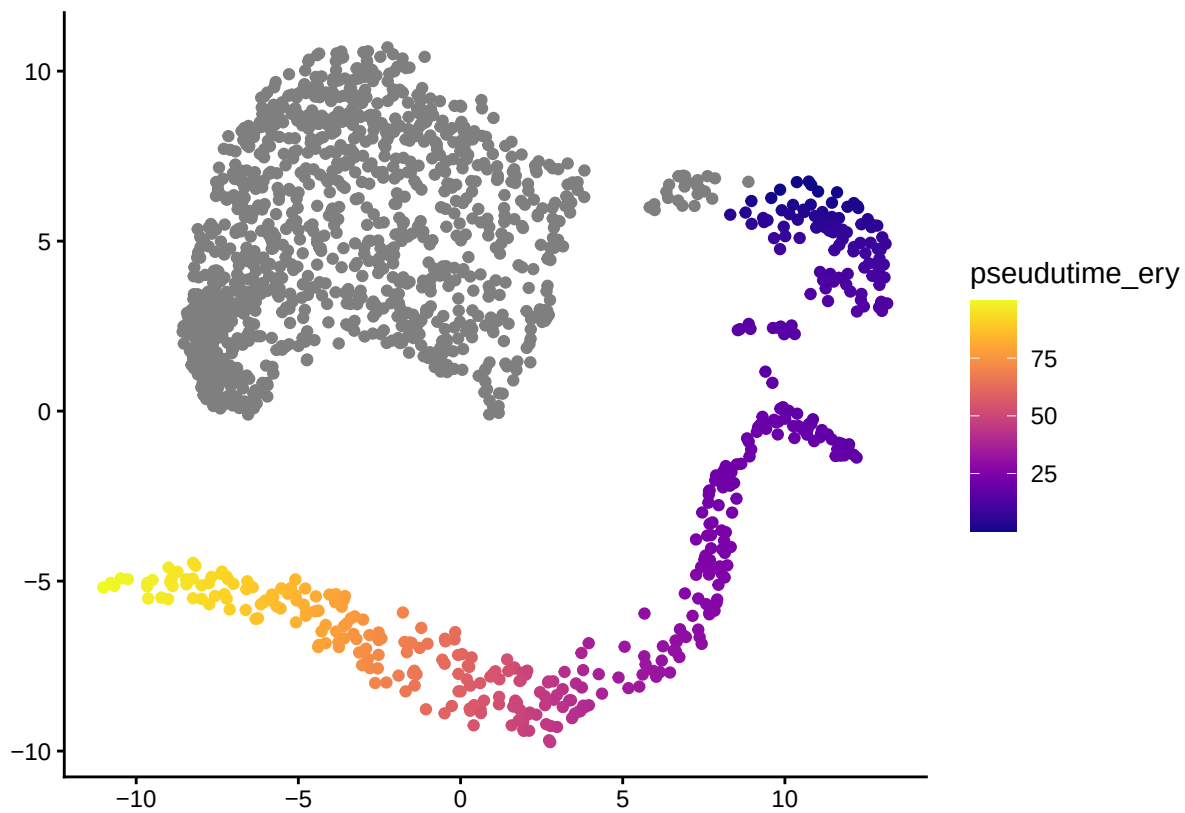
```
p_pseudotimeByGroupQunMy
## Warning: Removed 347 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



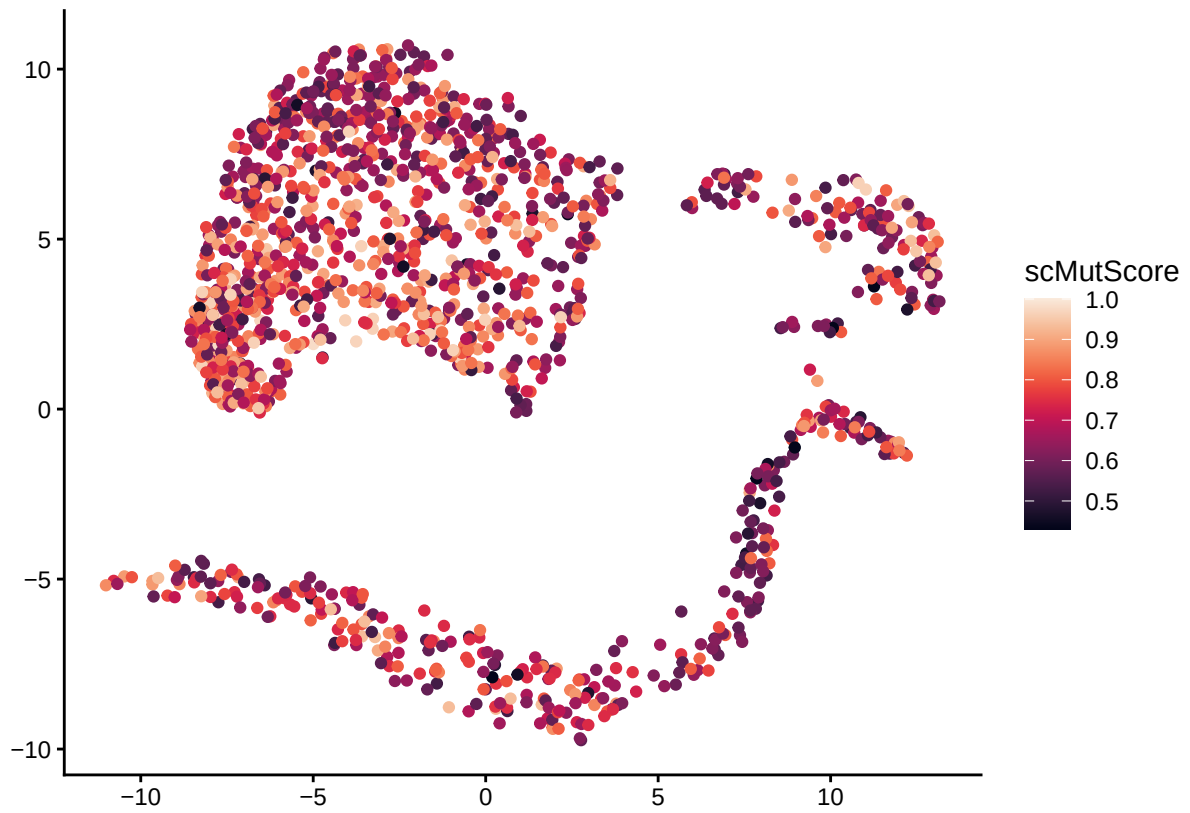
p_UMAP_pseudotime_my



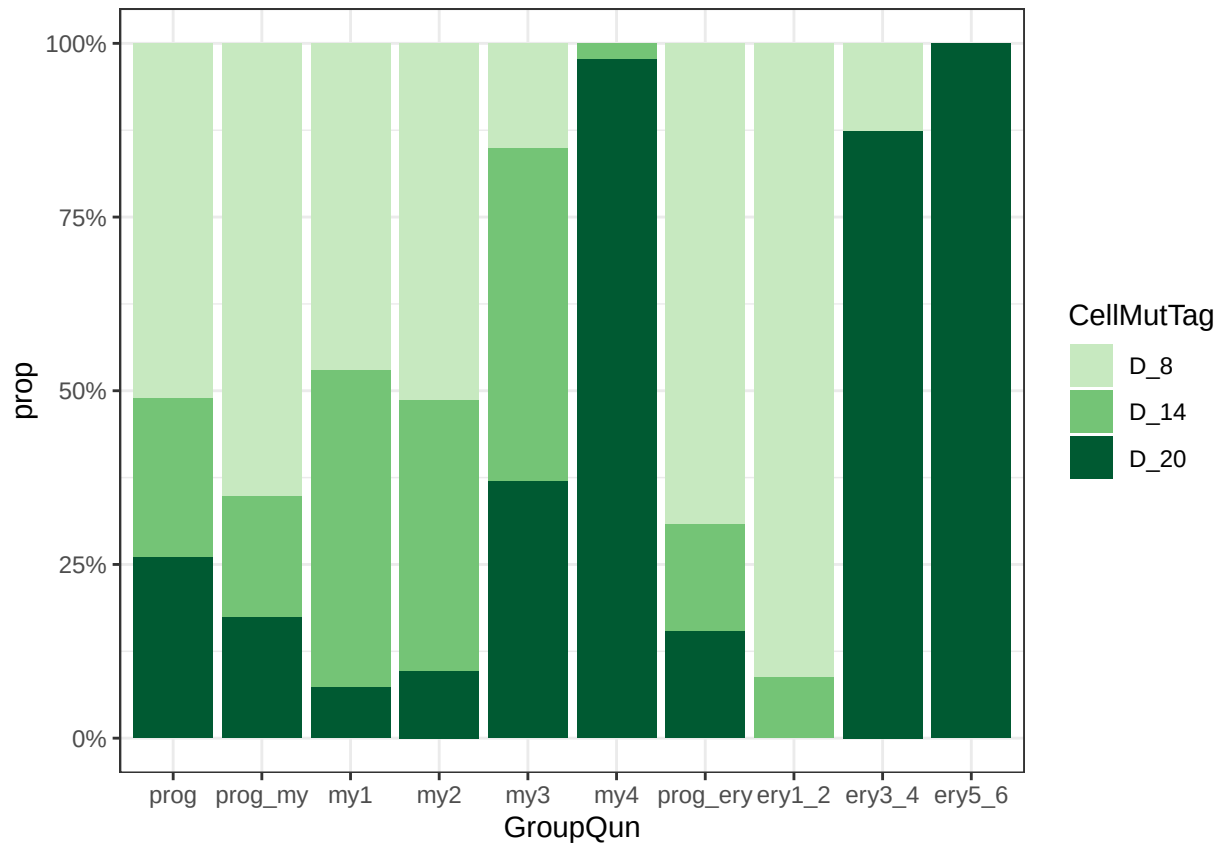
p_UMAP_pseudotime_ery



p_UMAP_scMutrace



p_cellfrac

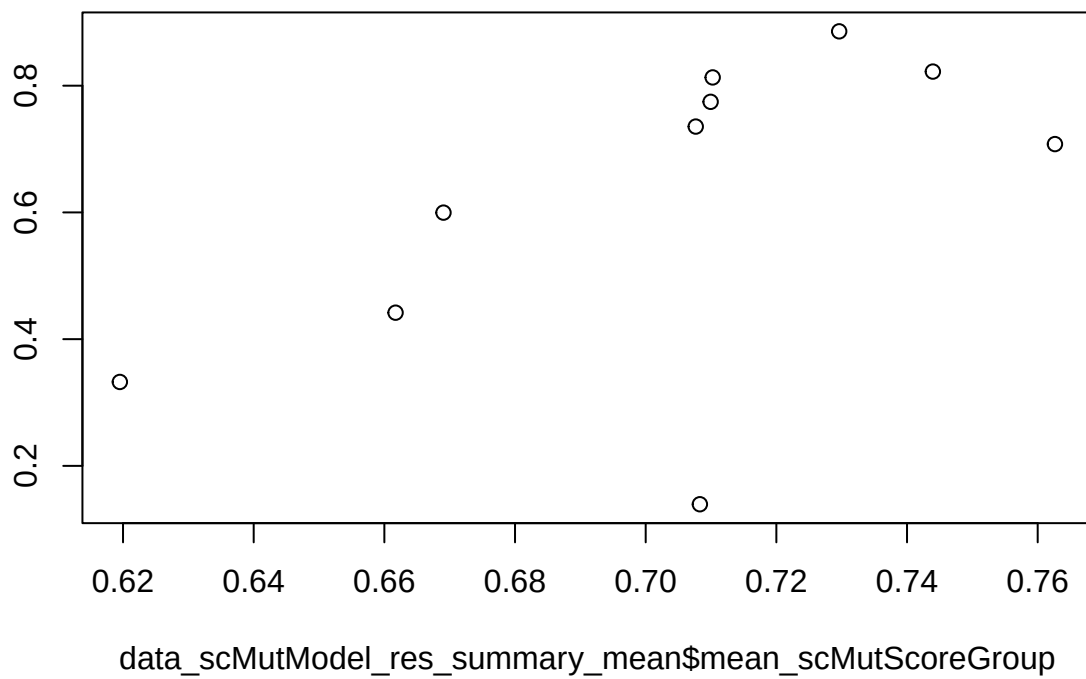


12 correlation of scMutrace and EpiAge

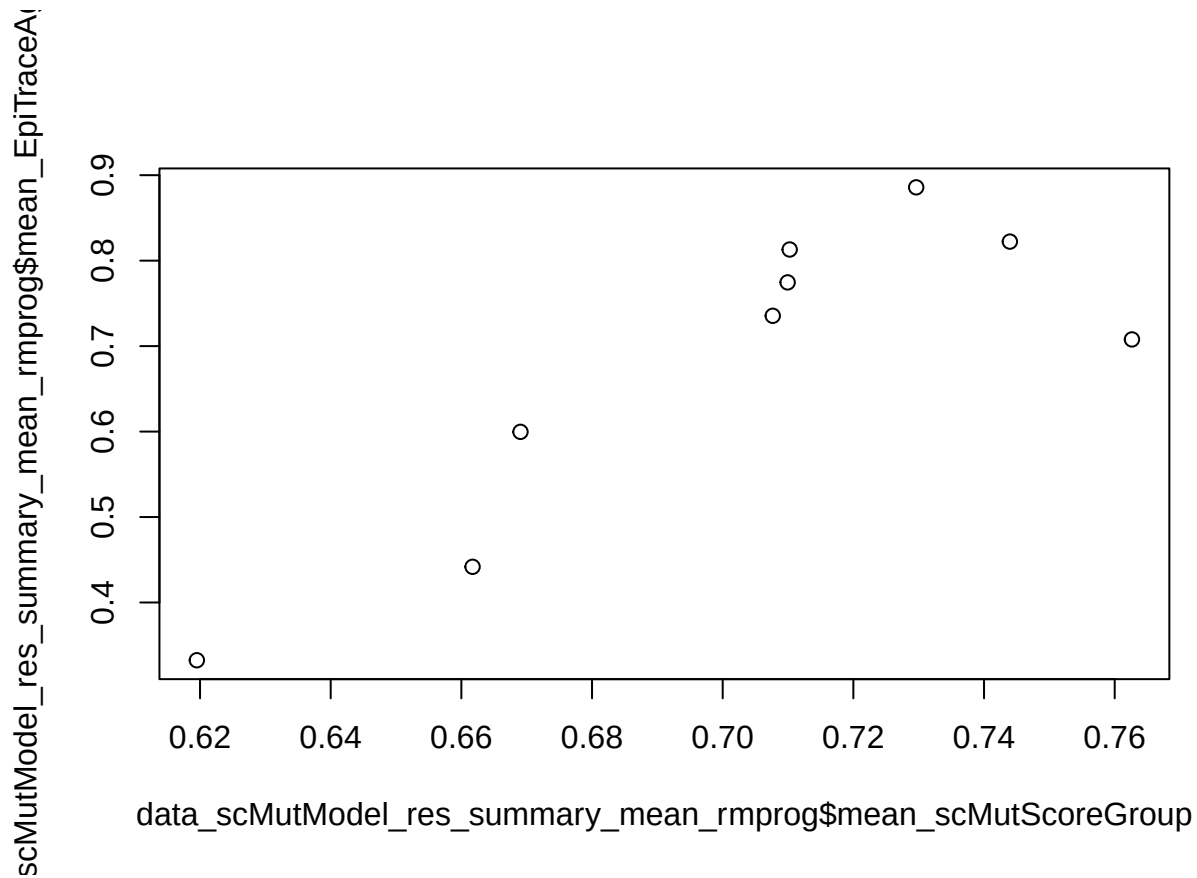
```
data_scMutModel_res_summary_mean <- data_scMutModel_res_summary %>%
  dplyr::group_by(GroupQun) %>%
  dplyr::summarise(mean_scMutScoreGroup = mean(mean_scMutScore, na.rm = TRUE), mean_EpiTraceAgeGroup = mean(EpiTraceAgeGroup),
  dplyr::ungroup()

data_scMutModel_res_summary_mean_rmprog <- data_scMutModel_res_summary_mean[data_scMutModel_res_summary$GroupQun != "prog", ]
plot(data_scMutModel_res_summary_mean$mean_scMutScoreGroup, data_scMutModel_res_summary_mean$mean_EpiTraceAgeGroup)
```

ata_scMutModel_res_summary_mean\$EpiTraceAgeG

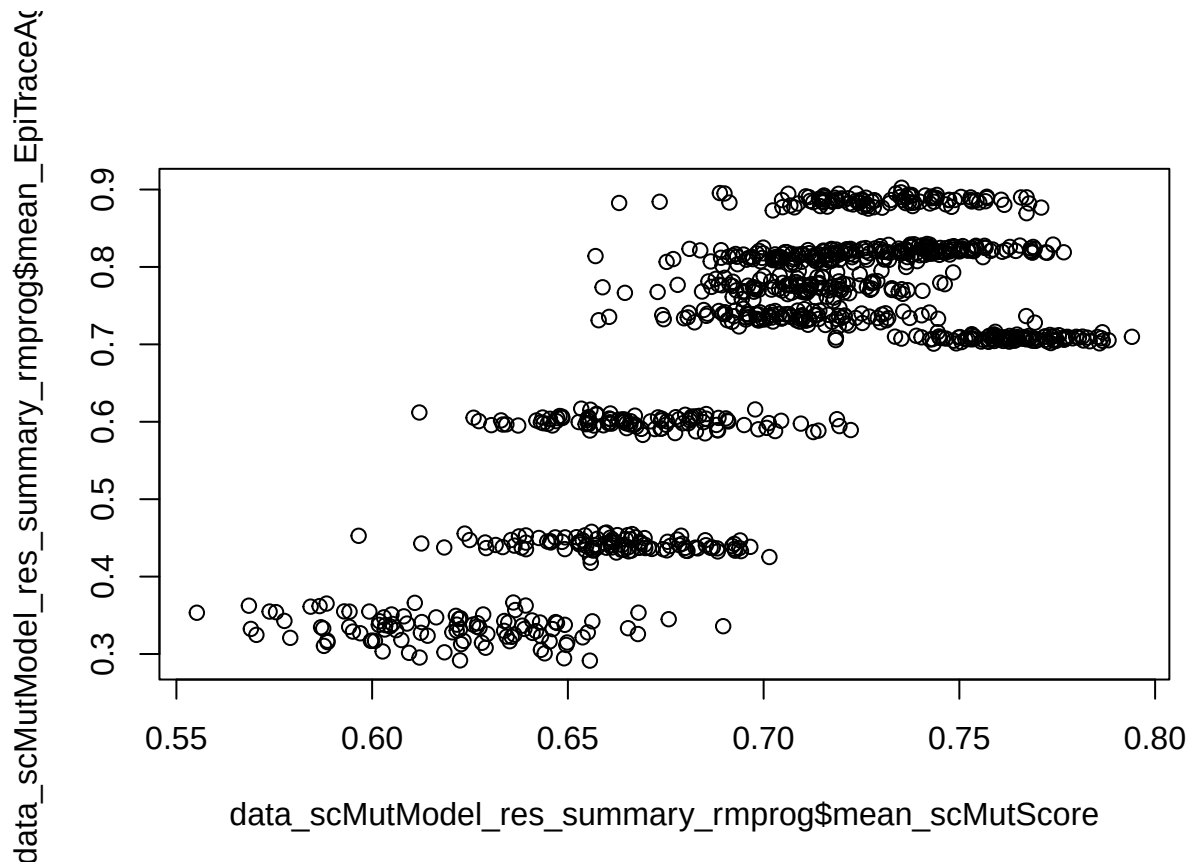


```
plot(data_scMutModel_res_summary_mean_rmprog$mean_scMutScoreGroup, data_scMutModel_res_summary_mean_rmp
```



```
cor.test(data_scMutModel_res_summary_mean$mean_scMutScoreGroup, data_scMutModel_res_summary_mean$mean_EpiTraceAge)
##
## Pearson's product-moment correlation
##
## data: data_scMutModel_res_summary_mean$mean_scMutScoreGroup and data_scMutModel_res_summary_mean$mean_EpiTraceAge
## t = 1.9816, df = 8, p-value = 0.08284
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.08740827 0.88403944
## sample estimates:
## cor
## 0.5737967
cor.test(data_scMutModel_res_summary_mean_rmprog$mean_scMutScoreGroup, data_scMutModel_res_summary_mean_rmprog$mean_EpiTraceAge)
##
## Pearson's product-moment correlation
##
## data: data_scMutModel_res_summary_mean_rmprog$mean_scMutScoreGroup and data_scMutModel_res_summary_mean_rmprog$mean_EpiTraceAge
## t = 4.2799, df = 7, p-value = 0.003656
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.4285777 0.9679331
## sample estimates:
## cor
## 0.8505962
cor.test(data_scMutModel_res_summary$mean_scMutScore, data_scMutModel_res_summary$mean_EpiTraceAge)
```

```
##
## Pearson's product-moment correlation
##
## data: data_scMutModel_res_summary$mean_scMutScore and data_scMutModel_res_summary$mean_EpiTraceAge
## t = 19.009, df = 998, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.4685630 0.5596839
## sample estimates:
## cor
## 0.5155797
data_scMutModel_res_summary_rmprog <- data_scMutModel_res_summary[data_scMutModel_res_summary$GroupQun
cor.test(data_scMutModel_res_summary_rmprog$mean_scMutScore, data_scMutModel_res_summary_rmprog$mean_Ep
##
## Pearson's product-moment correlation
##
## data: data_scMutModel_res_summary_rmprog$mean_scMutScore and data_scMutModel_res_summary_rmprog$mean
## t = 36.301, df = 898, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## 0.7432966 0.7963990
## sample estimates:
## cor
## 0.7711859
plot(data_scMutModel_res_summary_rmprog$mean_scMutScore, data_scMutModel_res_summary_rmprog$mean_EpiTraceAge)
```



13 savePic

```
pdf("../Figure/CD34/p_UMAP_Days.pdf", width = 4.47, height = 3.20)
p_UMAP_Days
dev.off()

pdf("../Figure/CD34/p_UMAP_Groups.pdf", width = 4.47, height = 3.20)
p_UMAP_Groups
dev.off()

pdf("../Figure/CD34/p_UMAP_pseudotime_my.pdf", width = 4.47, height = 3.20)
p_UMAP_pseudotime_my
dev.off()

pdf("../Figure/CD34/p_UMAP_pseudotime_ery.pdf", width = 4.47, height = 3.20)
p_UMAP_pseudotime_ery
dev.off()

pdf("../Figure/CD34/p_scMutScorebydays.pdf", width = 4.47, height = 3.20)
p_scMutScorebydays
dev.off()

pdf("../Figure/CD34/p_scMutScorebycelltypes.pdf", width = 4.47, height = 3.20)
p_scMutScorebycelltypesQun
dev.off()

pdf("../Figure/CD34/p_EpiScorebyCelltype.pdf", width = 4.47, height = 3.20)
p_EpiScorebyCelltypeQun
dev.off()

pdf("../Figure/CD34/p_pseudotimeByDaysEry.pdf", width = 4.47, height = 3.20)
p_pseudotimeByDaysEry
dev.off()

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p_pseudotimeByDaysMy
dev.off()

pdf("../Figure/CD34/p_pseudotimeByGroupQunEry.pdf", width = 4.47, height = 3.20)
p_pseudotimeByGroupQunEry
dev.off()

pdf("../Figure/CD34/p_pseudotimeByGroupQunMy.pdf", width = 4.47, height = 3.20)
p_pseudotimeByGroupQunMy
dev.off()

pdf("../Figure/CD34/p_cellfracbylineage.pdf", width = 4.47, height = 3.20)
p_cellfrac
dev.off()

pdf("../Figure/CD34/p_scMutrace_iter_Day.pdf", width = 4.47, height = 3.20)
p_scMutrace_iter_Day_scaled
dev.off()
```

```
pdf("../Figure/CD34/p_scMutrace_iter_GroupQun.pdf", width = 4.47, height = 3.20)
p_scMutrace_iter_GroupQunScaled
dev.off()

pdf("../Figure/CD34/p_iter_cellfrac.pdf", width = 4.47, height = 3.20)
p_iter_cellfrac
dev.off()

pdf("../Figure/CD34/p_randomMutNum_est.pdf", width = 4.47, height = 3.20)
p_randomMutNum_est
dev.off()
```

14 sessionInfo

```
sessionInfo()
## R version 4.5.1 (2025-06-13)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.4.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/Stockholm
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## other attached packages:
## [1] patchwork_1.3.2    lubridate_1.9.4    forcats_1.0.0      purrr_1.1.0
## [5] readr_2.1.5        tibble_3.3.0       tidyverse_2.0.0    networkD3_0.4.1
## [9] tidygraph_1.3.1    reshape2_1.4.4     pheatmap_1.0.13    corrplot_0.95
## [13] stringr_1.5.2      ggpmisc_0.6.2      ggpp_0.5.9         webshot_0.5.5
## [17] ggraph_2.2.2       igraph_2.1.4       Seurat_5.3.0       SeuratObject_5.2.0
## [21] sp_2.2-0           ggpubr_0.6.1       ggplot2_4.0.0      tidyr_1.3.1
## [25] dplyr_1.1.4        vegan_2.7-1        permute_0.9-8      ineq_0.2-13
## [29] FSA_0.10.0
##
## loaded via a namespace (and not attached):
## [1] RcppAnnoy_0.0.22    splines_4.5.1      later_1.4.4
## [4] polyclip_1.10-7     fastDummies_1.7.5  lifecycle_1.0.4
## [7] rstatix_0.7.2       globals_0.18.0     lattice_0.22-7
## [10] MASS_7.3-65         backports_1.5.0    magrittr_2.0.4
## [13] plotly_4.11.0       rmarkdown_2.30     yaml_2.3.10
## [16] httpuv_1.6.16       sctransform_0.4.2  spam_2.11-1
## [19] spatstat.sparse_3.1-0 reticulate_1.43.0  cowplot_1.2.0
```

```

## [22] pbapply_1.7-4      RColorBrewer_1.1-3  abind_1.4-8
## [25] Rtsne_0.17         tweenr_2.0.3        data.tree_1.2.0
## [28] ggrepel_0.9.6      irlba_2.3.5.1       listenv_0.9.1
## [31] spatstat.utils_3.2-0 MatrixModels_0.5-4  goftest_1.2-3
## [34] RSpectra_0.16-2    spatstat.random_3.4-2 fitdistrplus_1.2-4
## [37] parallelly_1.45.1  codetools_0.2-20    ggforce_0.5.0
## [40] tidyselect_1.2.1   farver_2.1.2        viridis_0.6.5
## [43] matrixStats_1.5.0  spatstat.explore_3.5-3 jsonlite_2.0.0
## [46] progressr_0.16.0   Formula_1.2-5       ggribes_0.5.7
## [49] survival_3.8-3     tools_4.5.1         ica_1.0-3
## [52] Rcpp_1.1.0         glue_1.8.0          gridExtra_2.3
## [55] xfun_0.53          mgcv_1.9-3          withr_3.0.2
## [58] fastmap_1.2.0      SparseM_1.84-2      digest_0.6.37
## [61] timechange_0.3.0   R6_2.6.1            mime_0.13
## [64] colorspace_2.1-2   scattermore_1.2     tensor_1.5.1
## [67] dichromat_2.0-0.1  spatstat.data_3.1-8 generics_0.1.4
## [70] data.table_1.17.8  graphlayouts_1.2.2  httr_1.4.7
## [73] htmlwidgets_1.6.4  uwot_0.2.3          pkgconfig_2.0.3
## [76] gtable_0.3.6       lmtest_0.9-40       S7_0.2.0
## [79] htmltools_0.5.8.1  carData_3.0-5       dotCall64_1.2
## [82] scales_1.4.0       png_0.1-8           spatstat.univar_3.1-4
## [85] knitr_1.50         rstudioapi_0.17.1   tzdb_0.5.0
## [88] nlme_3.1-168       zoo_1.8-14          cachem_1.1.0
## [91] KernSmooth_2.23-26 parallel_4.5.1      miniUI_0.1.2
## [94] pillar_1.11.1      grid_4.5.1          vctrs_0.6.5
## [97] RANN_2.6.2         promises_1.3.3      car_3.1-3
## [100] xtable_1.8-4       cluster_2.1.8.1     evaluate_1.0.5
## [103] cli_3.6.5          compiler_4.5.1      crayon_1.5.3
## [106] rlang_1.1.6        future.apply_1.20.0 ggsignif_0.6.4
## [109] labeling_0.4.3     plyr_1.8.9          stringi_1.8.7
## [112] viridisLite_0.4.2  deldir_2.0-4        lazyeval_0.2.2
## [115] spatstat.geom_3.6-0 quantreg_6.1        Matrix_1.7-4
## [118] RcppHNSW_0.6.0     hms_1.1.3           future_1.67.0
## [121] shiny_1.11.1       ROCR_1.0-11         broom_1.0.10
## [124] memoise_2.0.1      polynom_1.4-1

```